

The Hand

Department of Anatomy

Musculoskeletal System

The Upper Limb (5)

Second Part

Dr. Osman Ahmed Osman

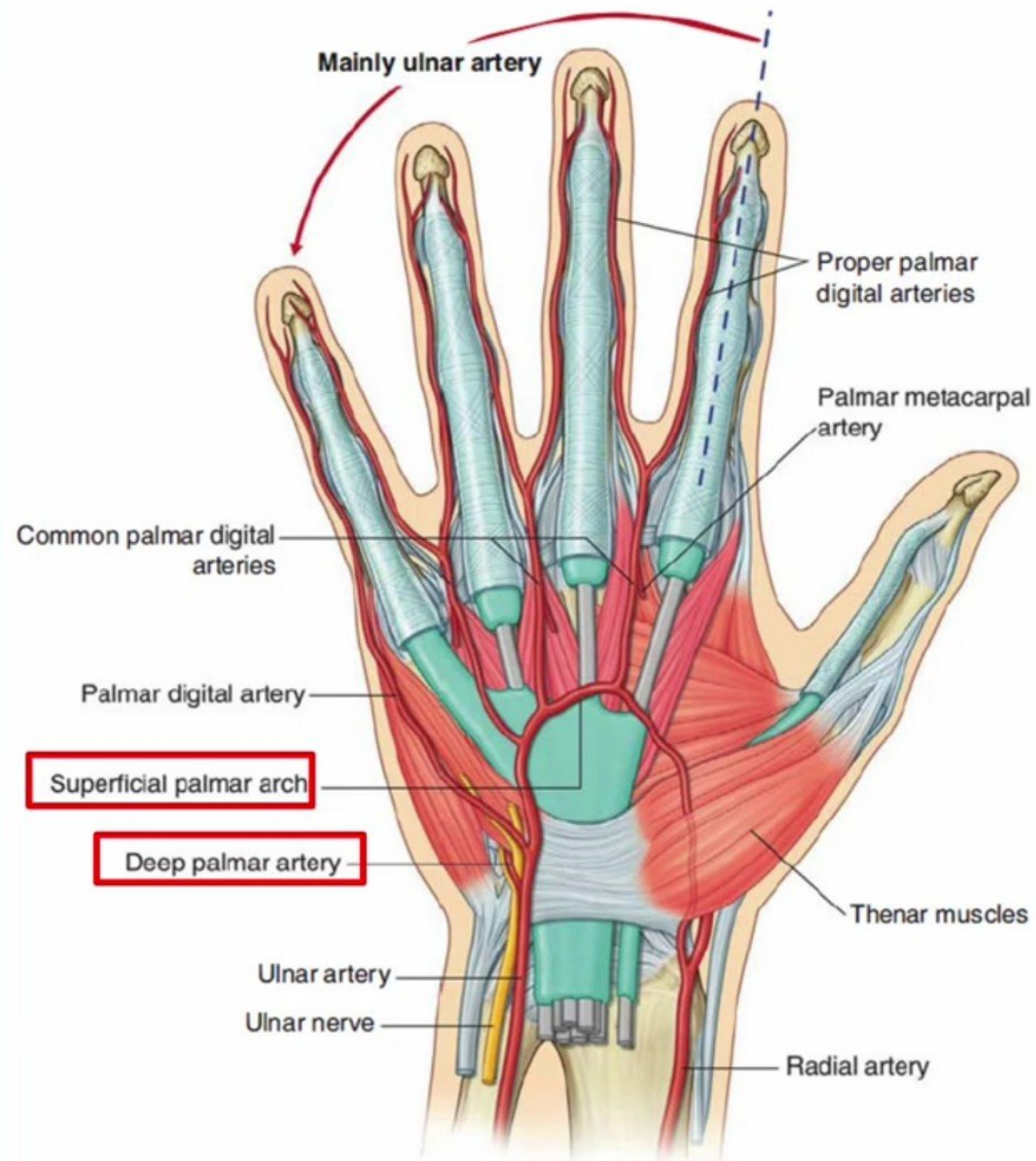
Lecture points

- Revision bone of the hand.
- Revision movement of the hand (Movements of the thumb).
- The Palmar aponeurosis
- Flexor retinaculum.
- The carpal tunnel.
- Muscles of the hand
 - Extrinsic Hand Muscles.
 - Intrinsic Hand Muscles.
 - Thenar muscles.
 - Hypothenar muscles.
 - Other.
- Blood supply
 - Arterial by (superficial and deep palmar arches).
 - The Dorsal venous arch of the hand.
- Lymph Drainage of the Palm
- Innervation of the hand.
- Fascial Palmer Spaces

The Arterial supply of Hand

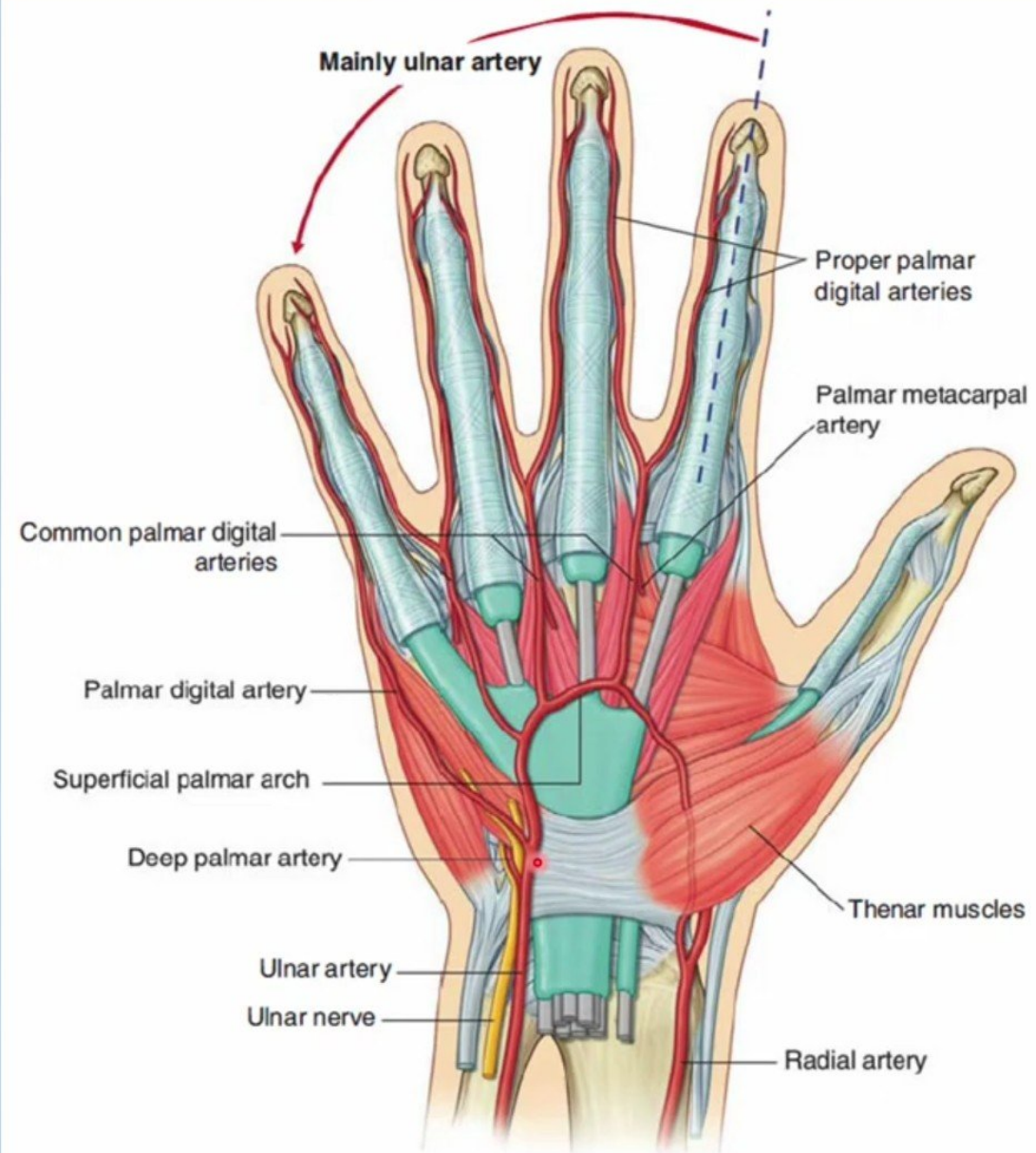
Ulnar Artery

- ✓ It enters the hand **above to the flexor retinaculum** on the lateral side of the **ulnar nerve** and the pisiform bone.
- ✓ It gives off a **deep palmar (branch)** and then **continues** into the palm as the **superficial palmar arch**.
- ✓ The **deep palmar branch of the ulnar artery** arises **in front** of the flexor retinaculum, passes between the **abductor digiti minimi** and the **flexor digiti minimi**, and joins **radial artery** to **complete** the **deep palmar arch**.



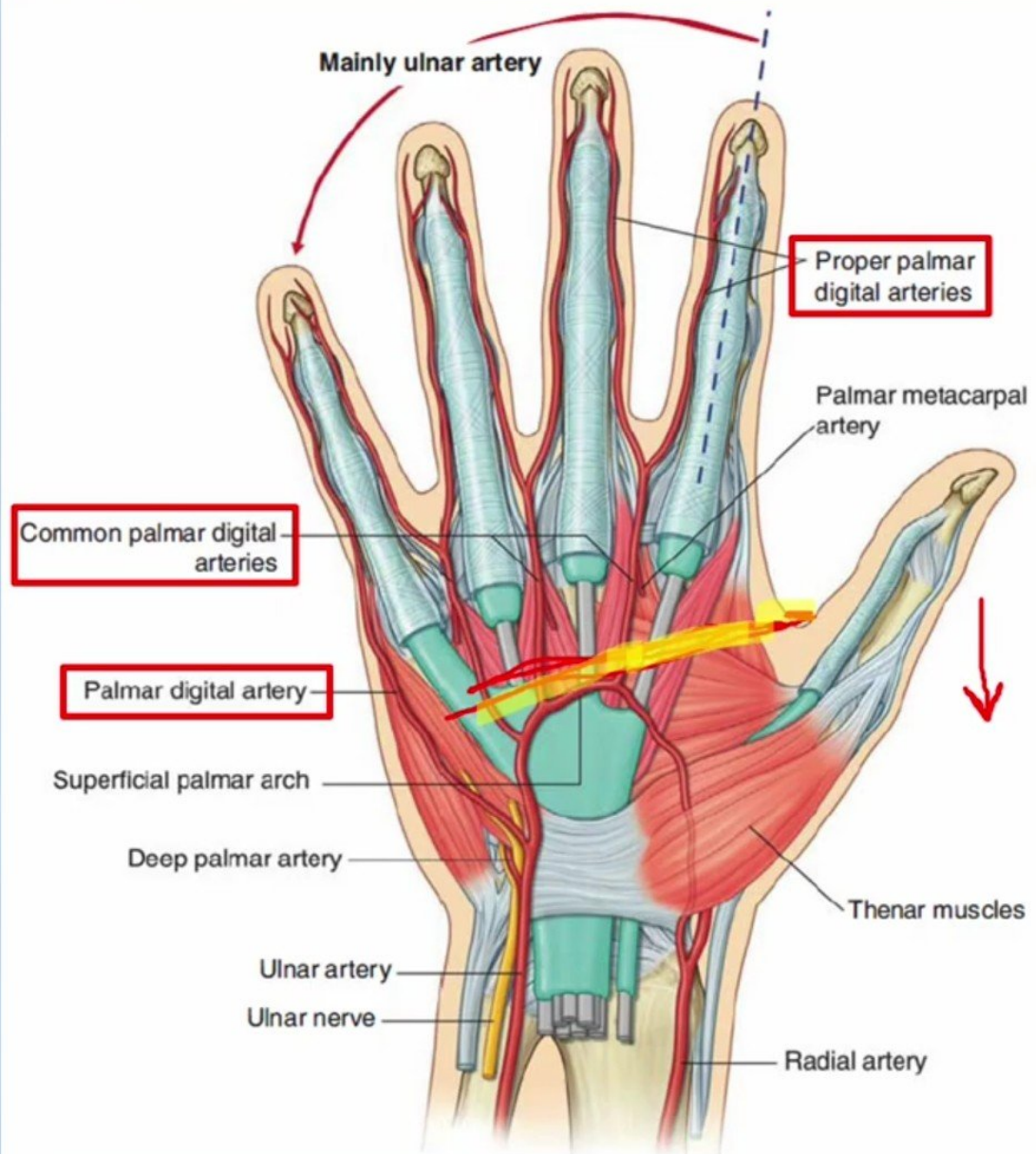
Superficial palmar arch

- This is an **arterial arcade**.
- It lies **superficial** to everything in **the palmar compartment**, (in contact with the deep surface of the palmar aponeurosis).
- It is formed by **the direct continuation of the ulnar artery** beyond the flexor retinaculum.
- It is **often not a complete arch**.
- If it is **complete** it **becomes continuous with the superficial palmar branch of the radial artery**.
- The arch lies **across the center of the palm**, (level with the distal border of the outstretched thumb).



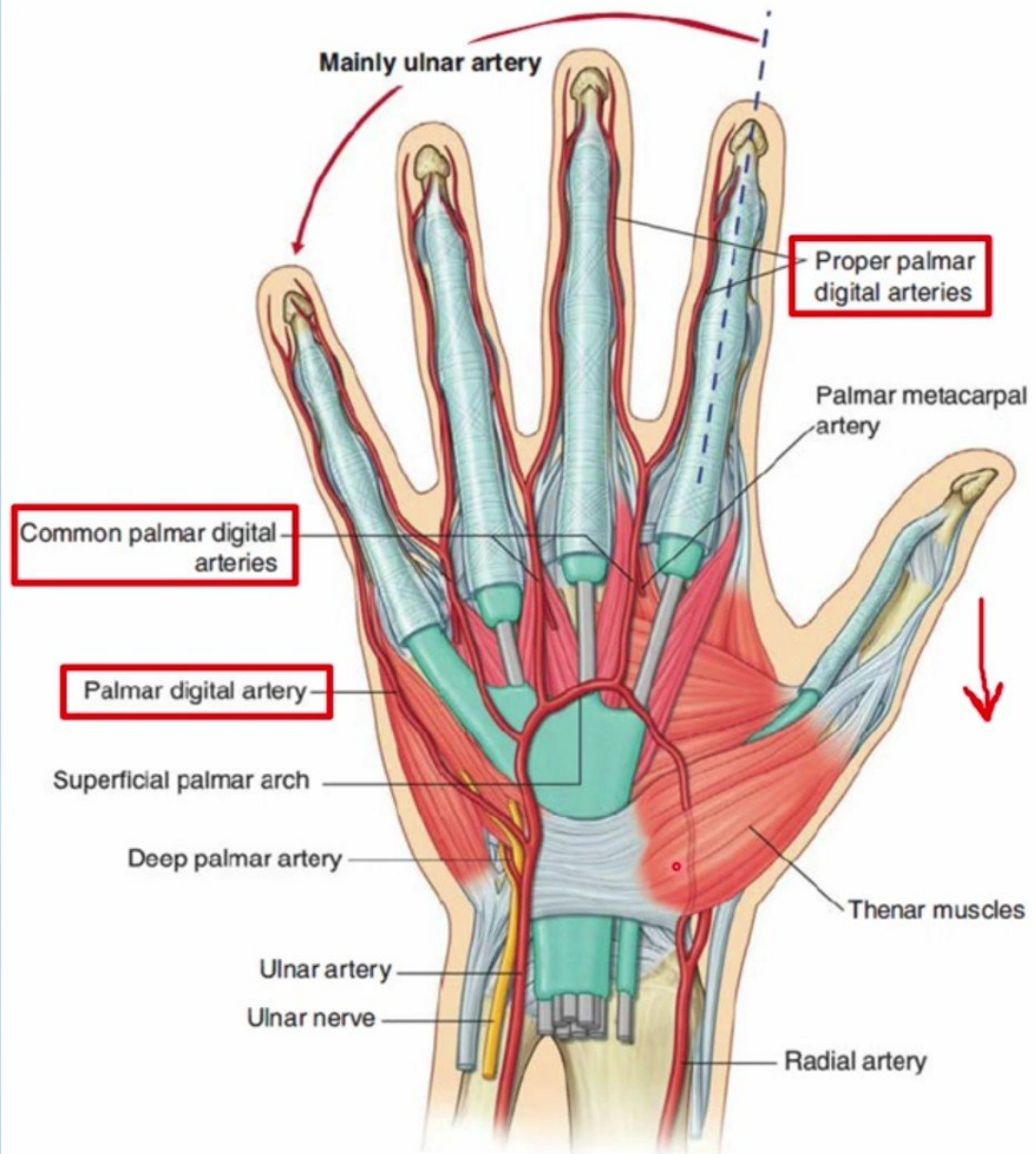
Superficial palmar arch

- From its convexity a palmar digital artery passes to the **ulnar side of the little finger**, and **three common palmar digital arteries** run distally to the webs between the fingers, where each vessel divides into proper palmar digital arteries that supply adjacent fingers.
- The thumb side of the index finger and the thumb itself are not supplied from **the superficial arch** since they **receive branches from the radial artery**.
- Palmar digital arteries anastomose with **dorsal digital arteries** and supply the **distal soft parts on the dorsum**, including the **nail beds**.



Superficial palmar arch

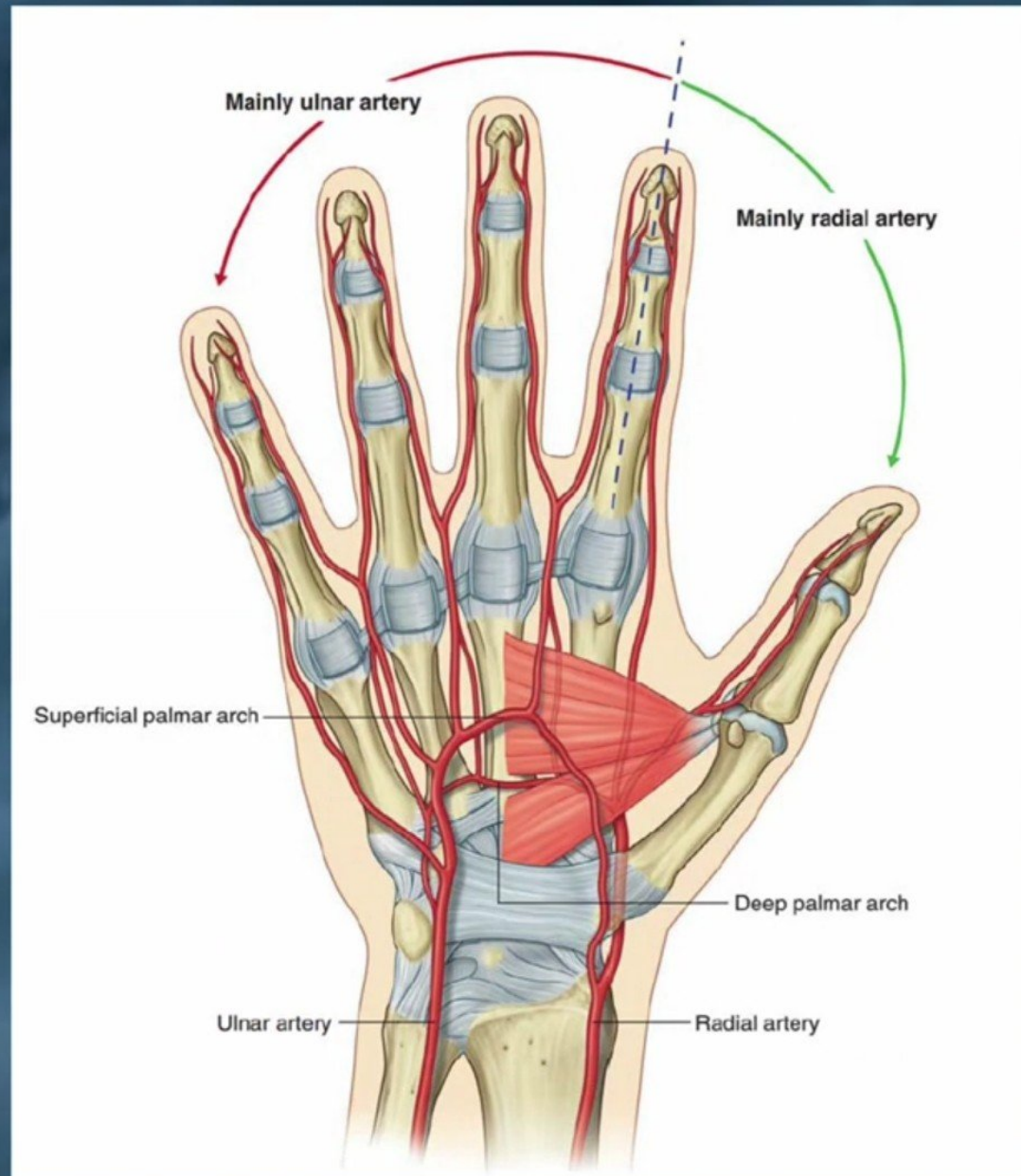
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The Arterial supply of Hand

Radial Artery

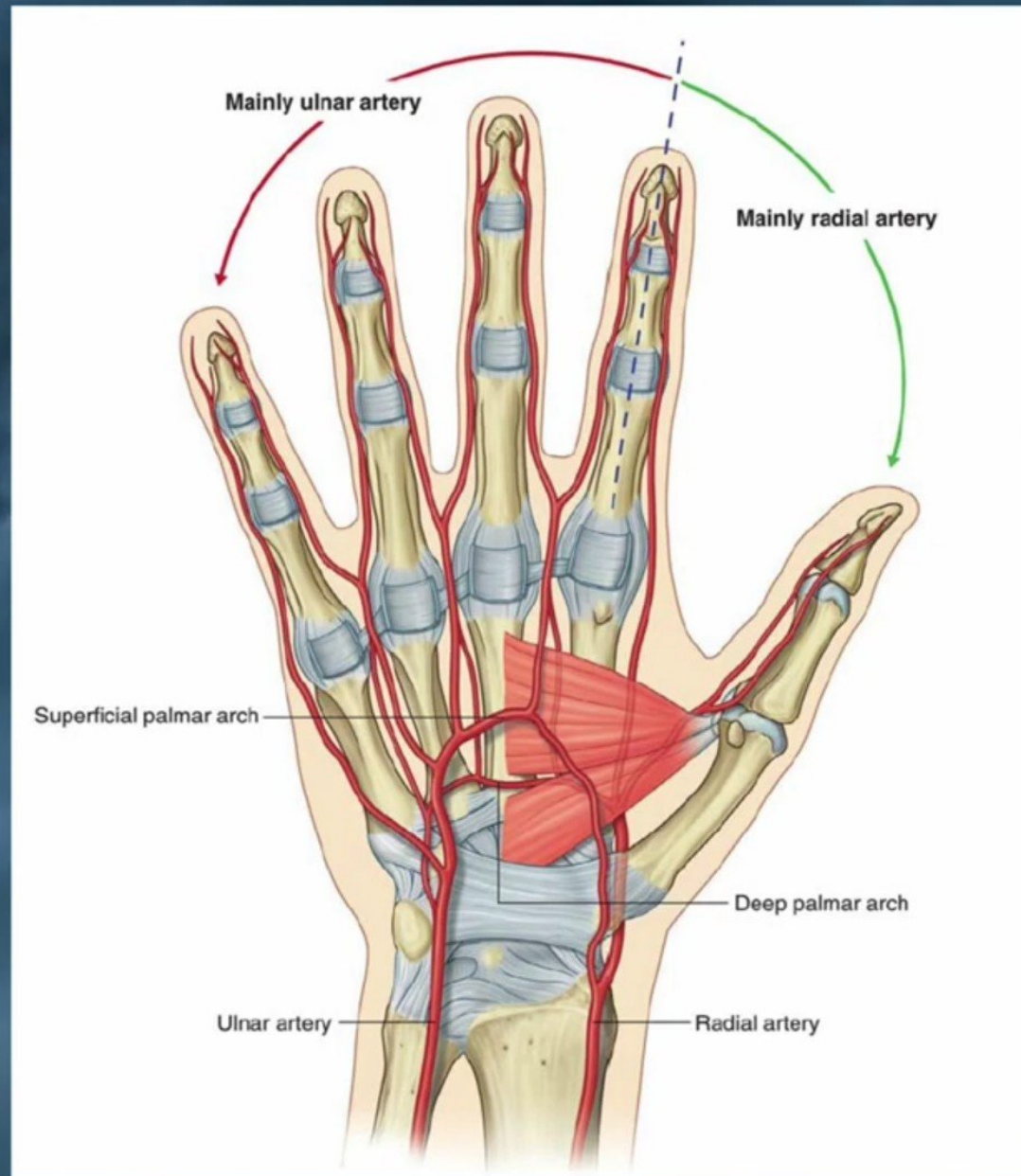
- ✓ It crosses the **anatomical snuffbox** over the **trapezium** and passes to the **dorsum of the hand**.
- ✓ It leaves the **dorsum of the hand** by turning forward between the proximal ends of **the first** and **second metacarpal bones** and **the two heads of the first dorsal interosseous muscle**.
- ✓ On entering the palm, it **curves medially** between **the oblique** and **transverse heads of the adductor pollicis** and **continues as the deep palmar arch**.



The Arterial supply of Hand

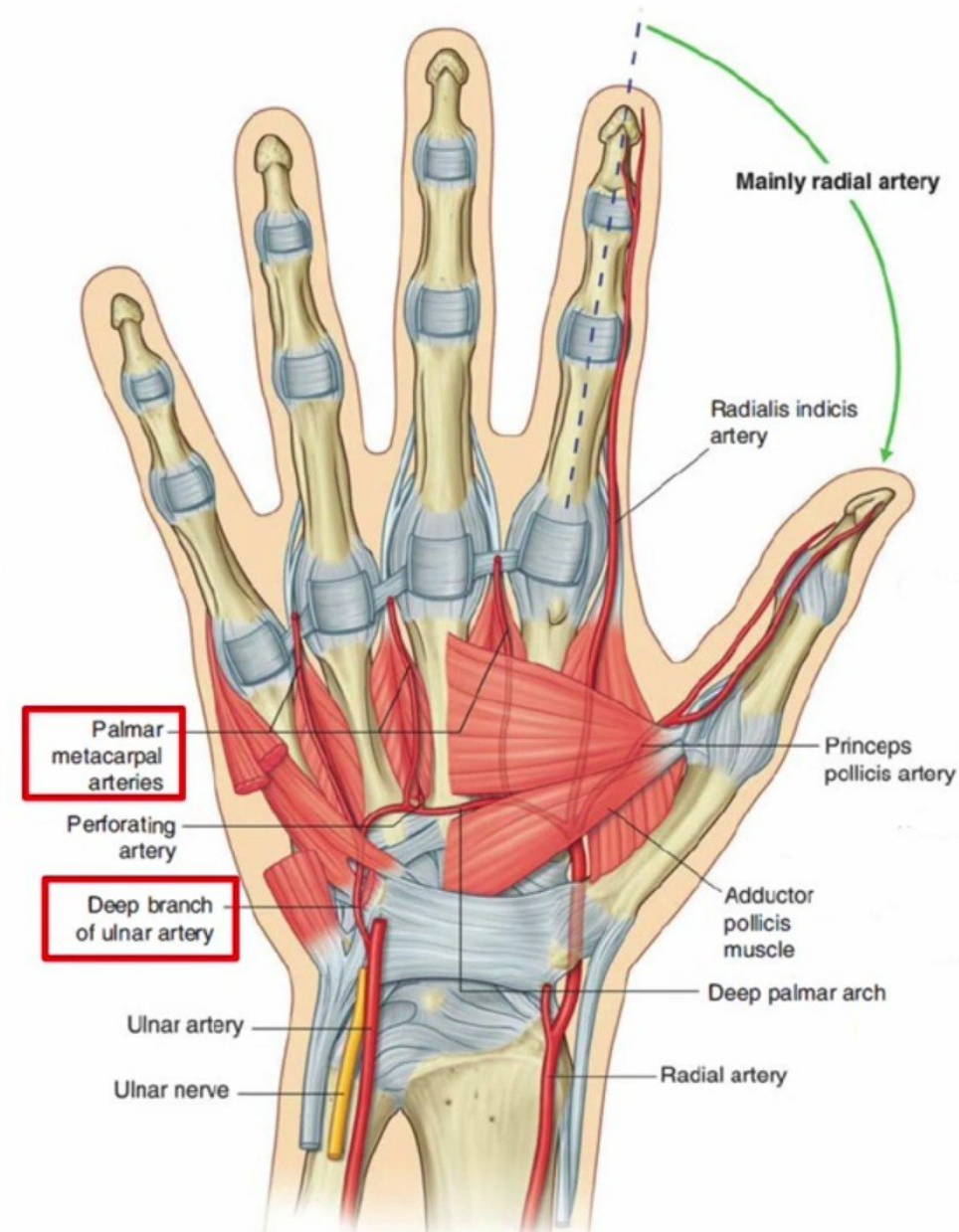
Radial Artery

- ✓ The deep palmar arch is a **direct continuation** of the radial artery.
- ✓ It curves medially beneath **the long flexor tendons** and in front of **the metacarpal bones** and **the interosseous muscles**.
- ✓ The arch is completed on the medial side by the **deep palmar branch** of the **ulnar artery**.
- ✓ The curve of the arch **lies at a level** with the proximal border of the **extended thumb**.



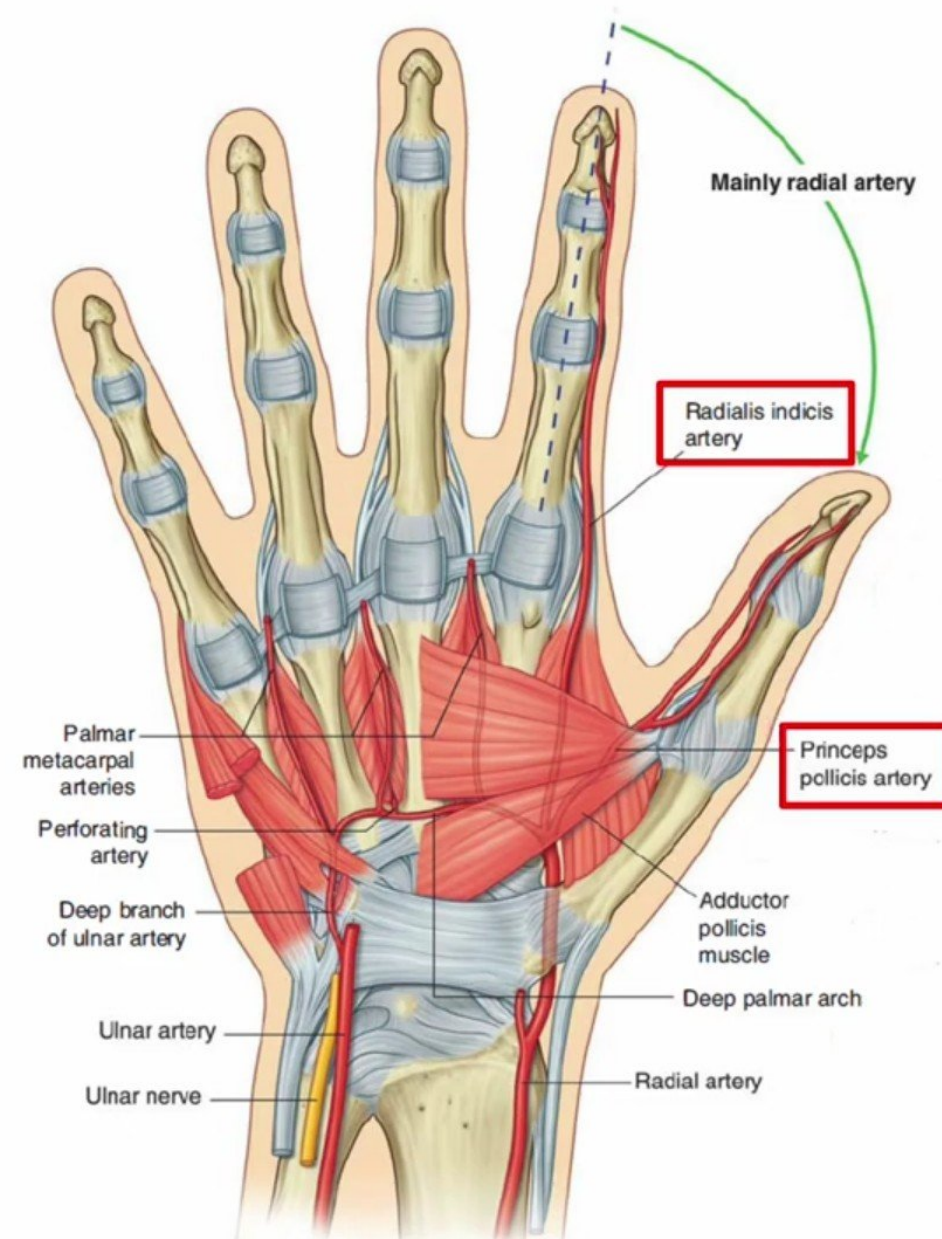
Deep palmar arch

- ✓ It is an **arterial arcade**.
- ✓ It is formed by **the terminal branch** of the **radial artery** anastomosing with **the deep palmar branch of the ulnar artery**.
- ✓ **The deep palmar branch of the ulnar nerve:**
 - It lies within the **concavity of the deep arch**.
 - From its convexity **three palmar metacarpal arteries** pass distally and in the region of **the metacarpal heads** they anastomose with the **common palmar digital branches** of the **superficial arch**.
 - **Branches perforate** pass between the interosseous spaces to anastomose with **the dorsal metacarpal arteries**.



Deep palmar arch

- ✓ Two vessels, **the princeps pollicis artery** and **the radialis indicis artery**.
 - It arise from **the radial artery** in the plane between **the first dorsal interosseous** and **adductor pollicis**.
 - **The princeps pollicis artery** is the **major blood supplies to the thumb**,
 - **The radialis indicis artery** supplies **the lateral side of the index finger**.



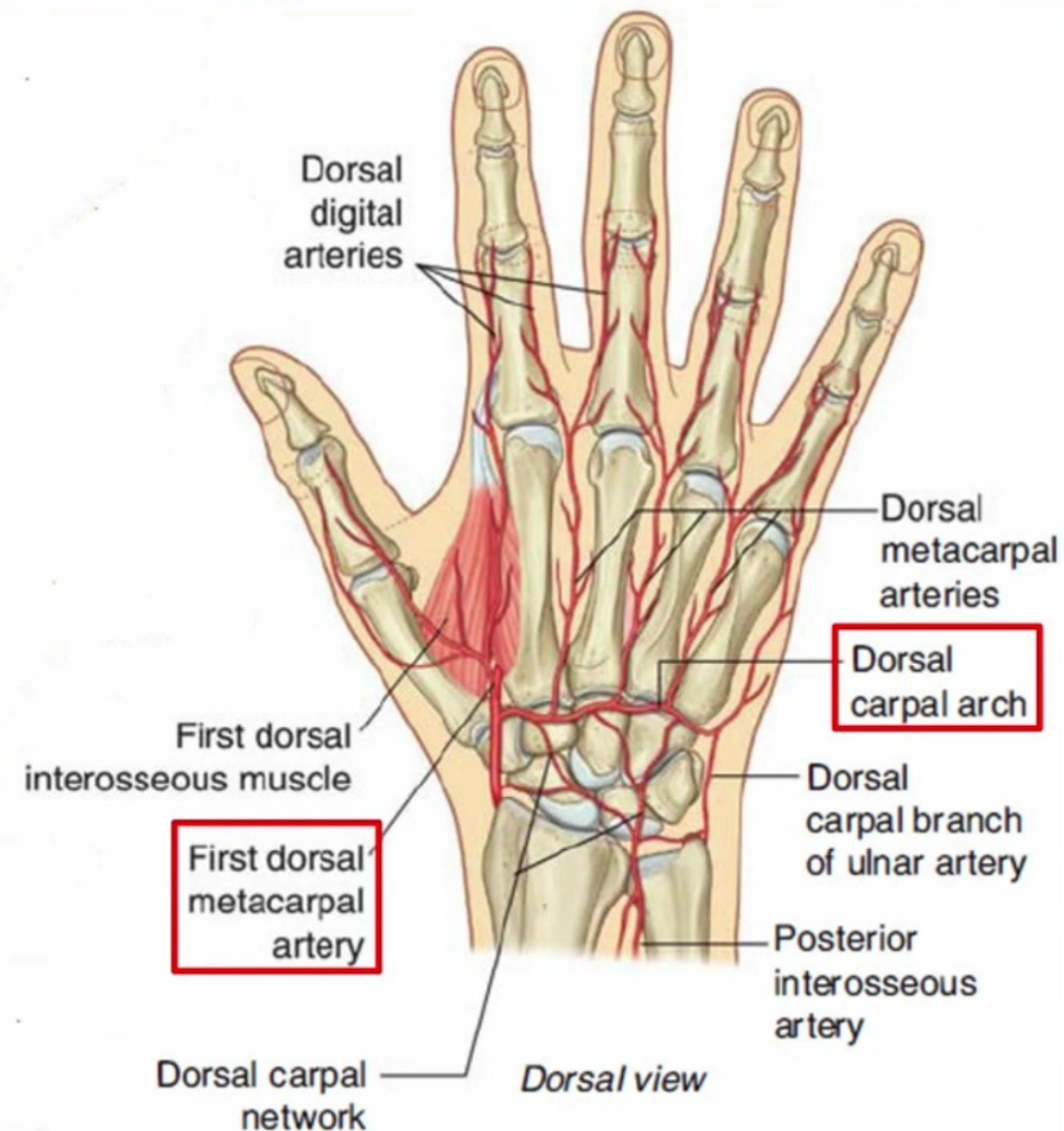
Deep palmar arch

✓ The dorsal carpal branch:

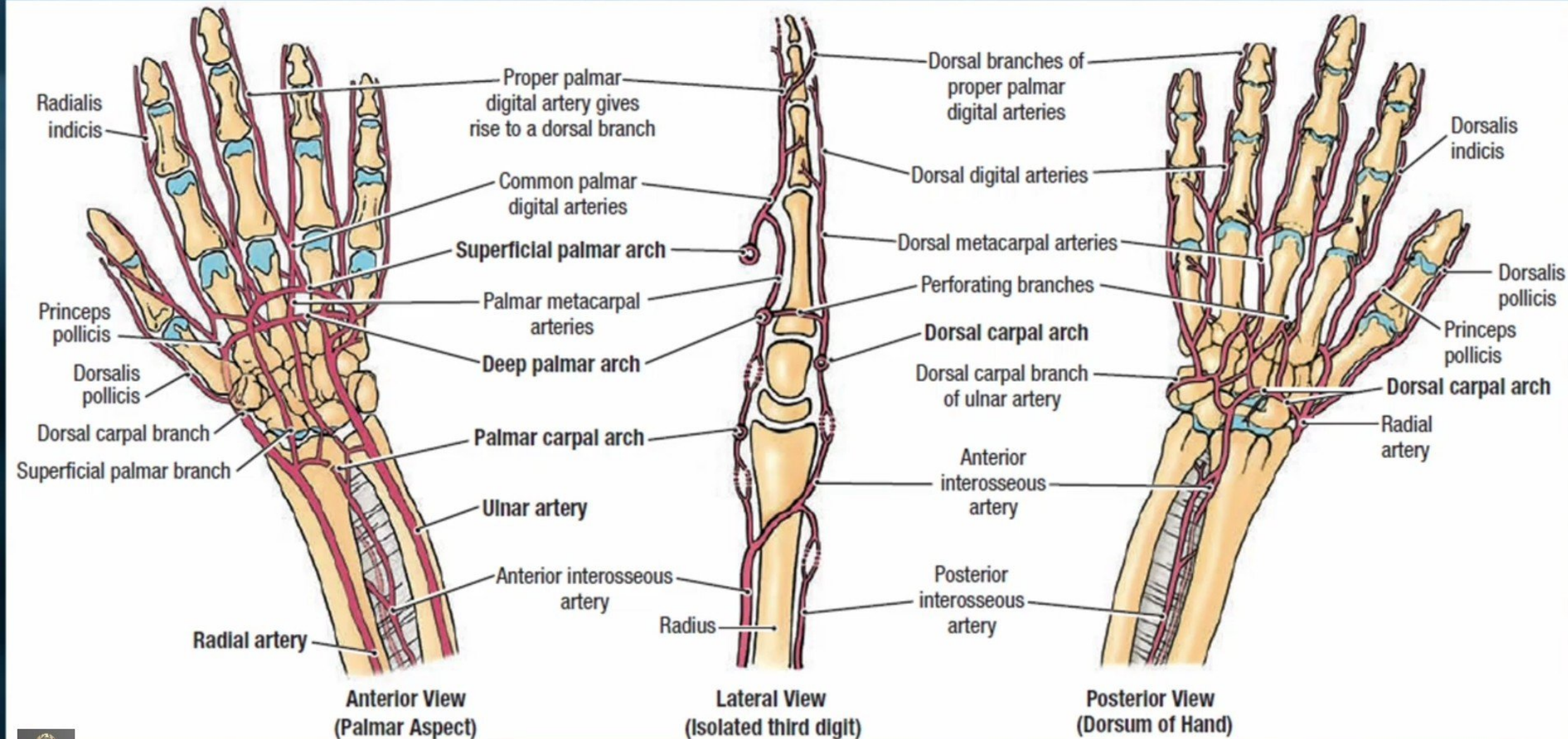
- It which passes **medially**, across the wrist.
- It gives rise to **three dorsal metacarpal arteries**, which subsequently divide to become **small dorsal digital arteries**, which **enter the fingers**.

✓ The first dorsal metacarpal artery:

- It which supplies adjacent sides of **the index fingers** and **thumbs**.

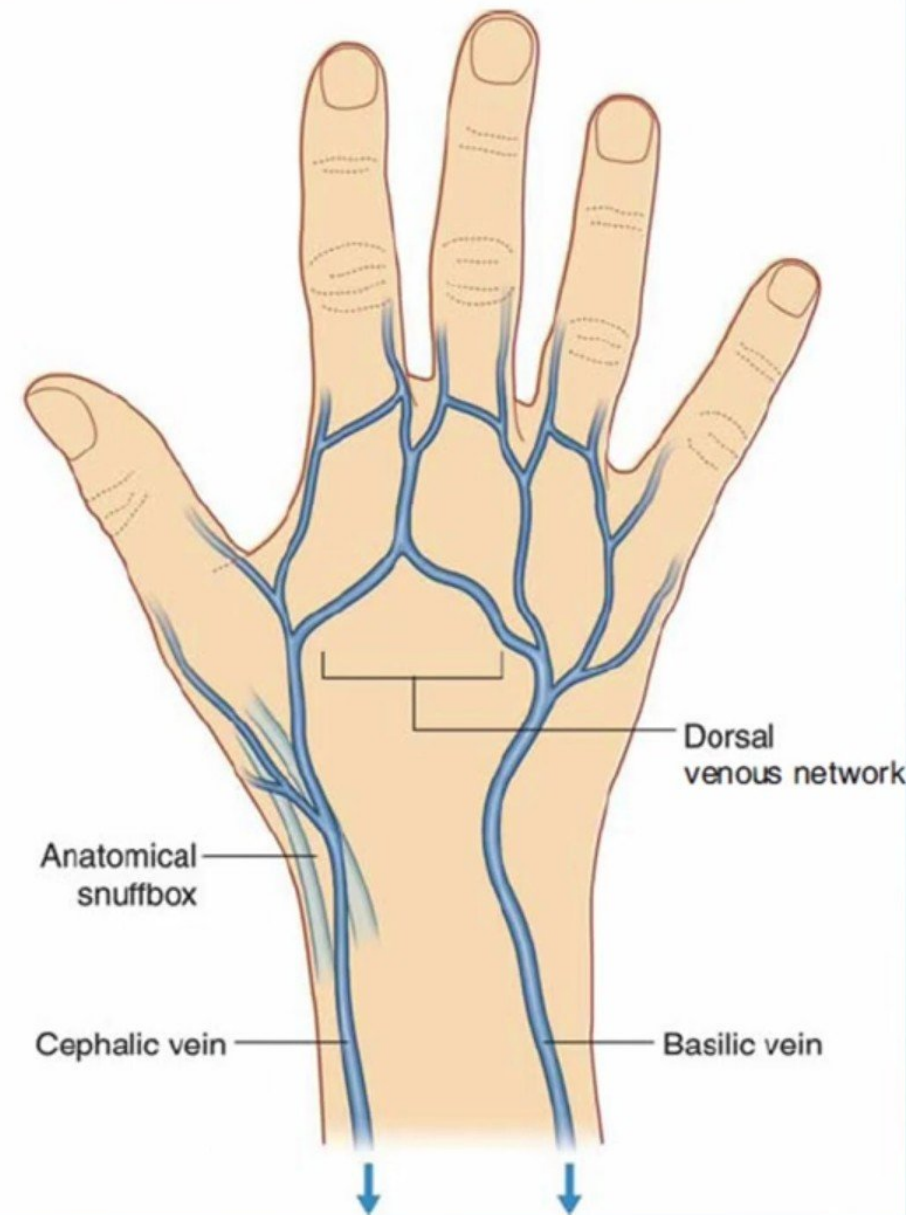


Summary of arterial supply of the hand



The Dorsal venous arch of the hand

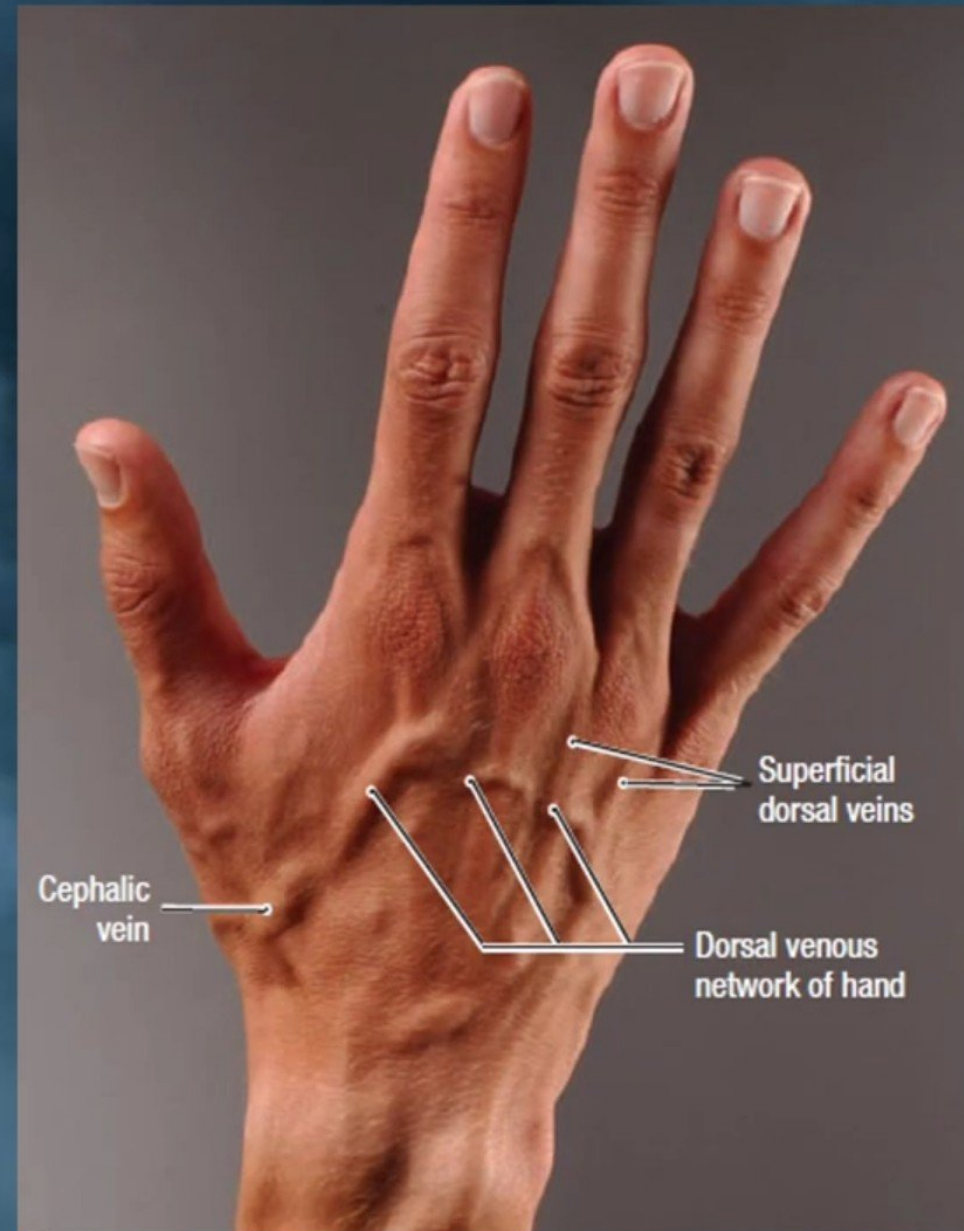
- Large veins forming the dorsal venous network lie **beneath** the **skin**; they drain from the palm, so that the pressure of gripping does not impede venous return.
- The network lies **superficial to the extensor tendons**, proximal to the **metacarpal heads**, and
- It drains on the radial side (laterally) into the cephalic vein and on the ulnar side (medially) into the basilic vein.



The Dorsal venous arch of the hand

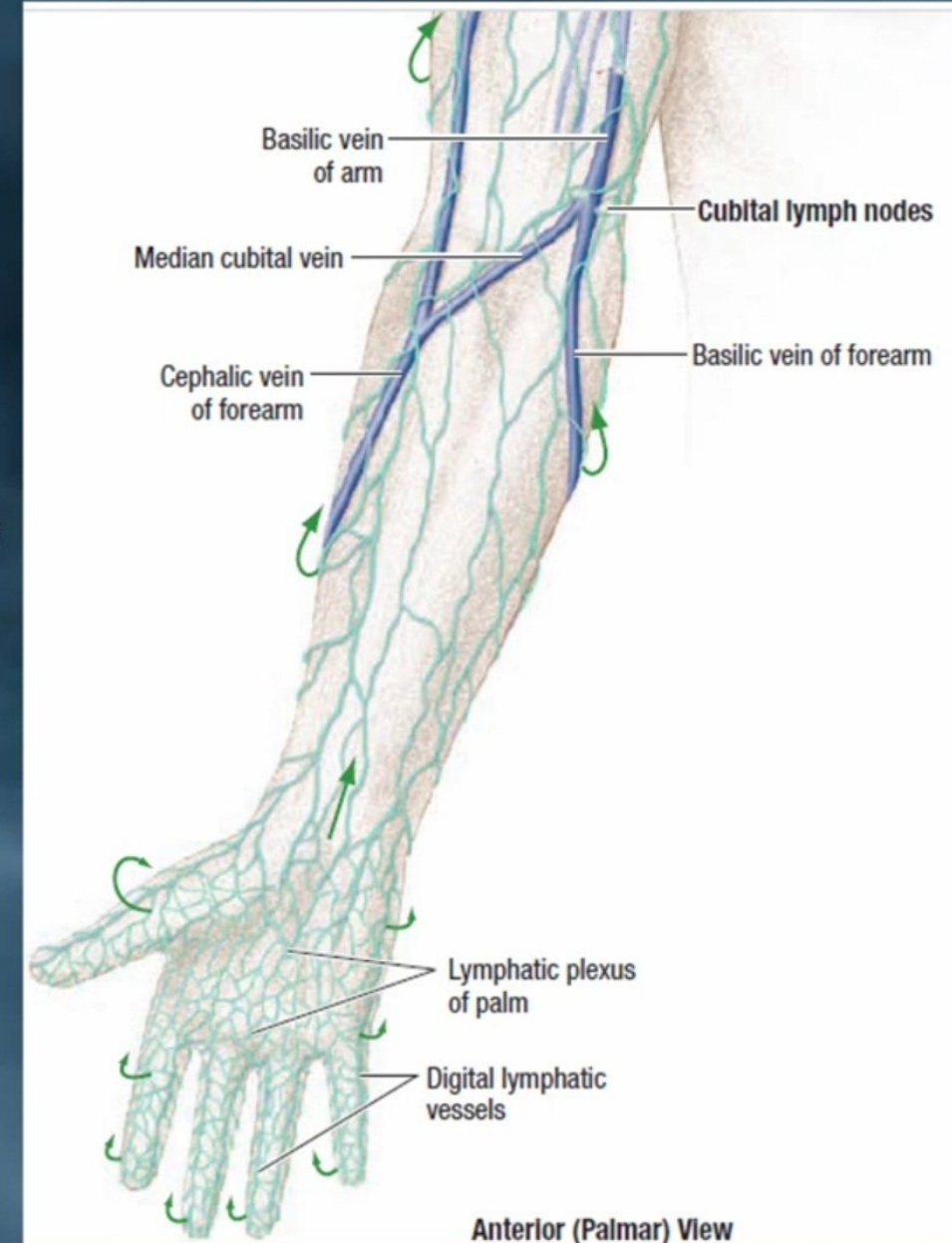
The skin of the dorsum:

- It is **thin** (little subcutaneous fat)
- It can be **picked up from the underlying deep fascia and tendons** (It moved freely over).



Lymph Drainage of the Palm

- The **lymph vessels of the fingers** pass **along their borders** to reach the webs. From here, the **vessels ascend onto the dorsum of the hand**.
- **Lymph vessels on the palm** form a **plexus** that is drained by vessels that **ascend in front of the forearm** or pass around the **medial and lateral** borders to join vessels on **the dorsum of the hand**.
- **The lymph** from **the medial side of the hand** ascends in vessels that **accompany the basilic vein**; they **drain into** the **supratrochlear nodes** and then ascend **to drain into** the **lateral axillary nodes**.
- **The lymph** from **the lateral side of the hand** ascends in vessels that **accompany the cephalic vein**; they **drain into** the **infraclavicular nodes**, and some drain into the **lateral axillary nodes**.



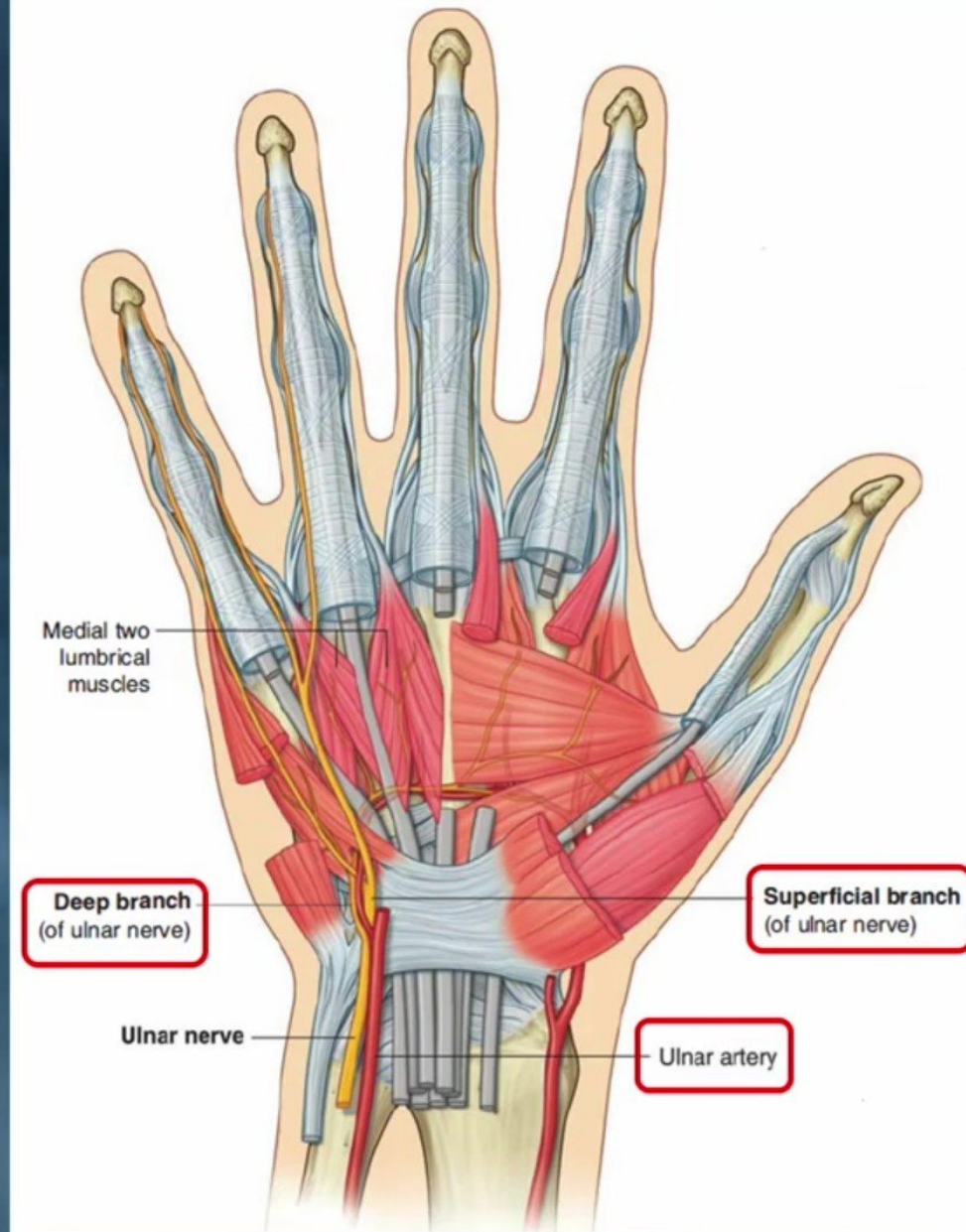
Innervation of the hand

The Ulnar innervation in the hand

- It leaves the forearm by emerging from **deep** to the tendon of **flexor carpi ulnaris**.
- It enters the palm **anterior** to **the flexor retinaculum** alongside **the lateral border of the pisiform bone** and **medial to the ulnar artery**.
- It divides into:
 - 1) **Superficial branch.**
 - 2) **Deep terminal branch.**

The tunnel of Guyon:

- It created by **fibrous tissue** derived **from the superficial part of the flexor retinaculum**.
- The **ulnar nerve** and **ulnar artery** may lie in a **fibro-osseous tunnel**.
- The nerve may be **compressed** at this site, giving rise to clinical signs and symptoms.



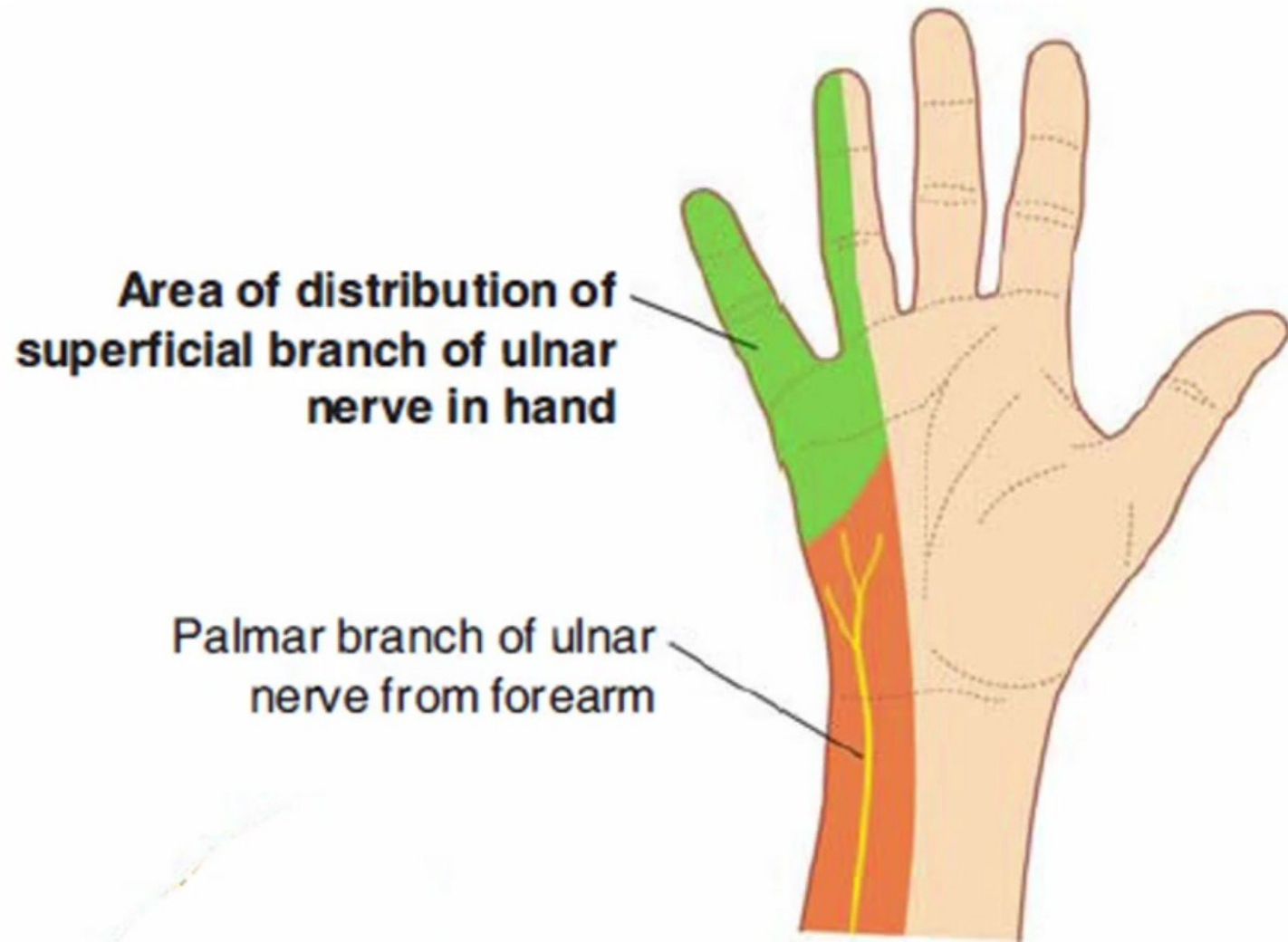
1) Superficial branch of the ulnar nerve

The superficial branch:

- It pass between **the pisiform bone** and **the hook of the hamate**.
- It **supplies palmaris brevis**.
- It divides into:
 - ✓ **Two digital nerves**; they supply **the ulnar one** and a **half fingers**.

The palmar cutaneous branch of the ulnar nerve

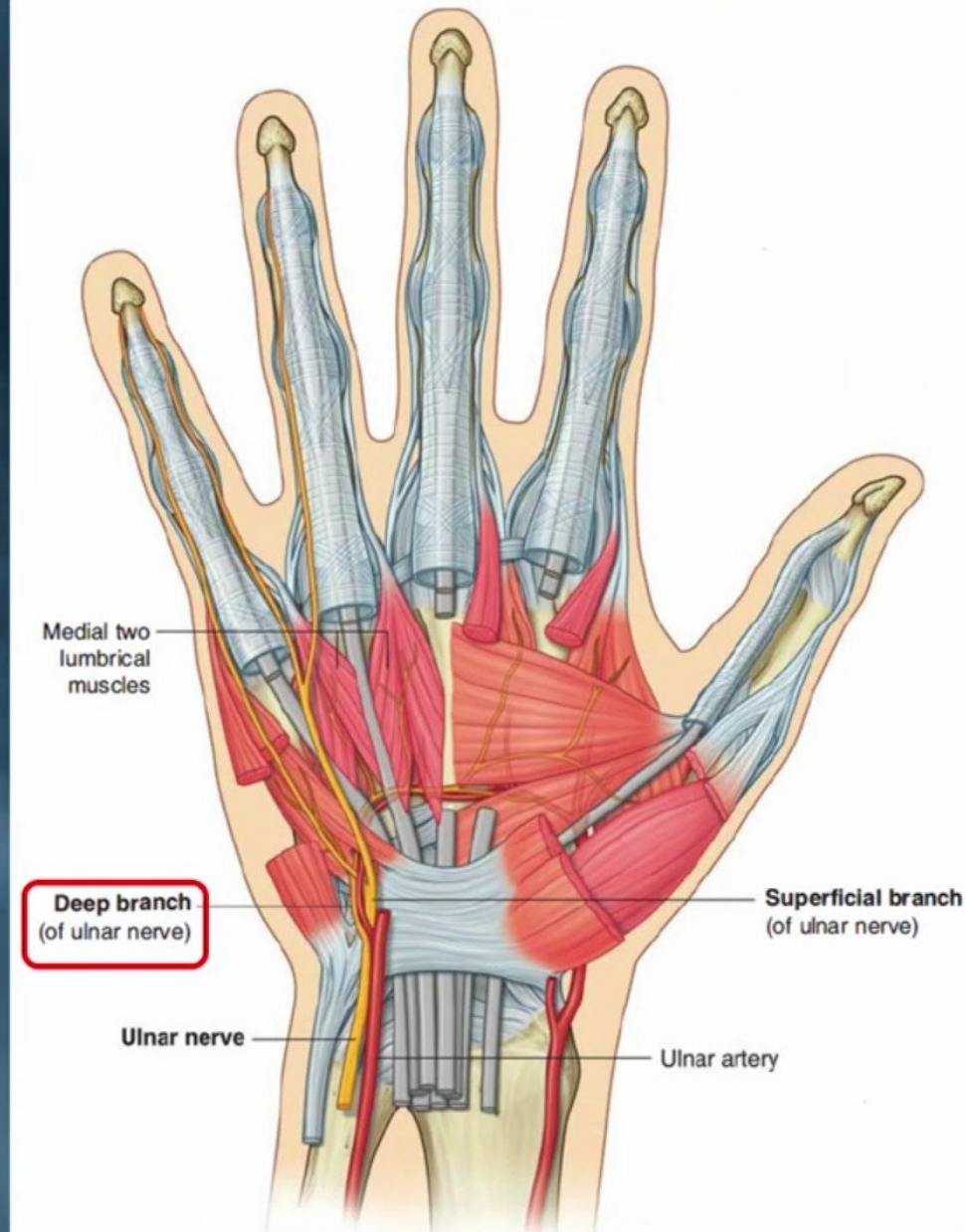
- The small palmar branch.
- It given off by the **ulnar nerve** in the **forearm**, supplies **skin over the hypothenar muscles**.



Palmar view

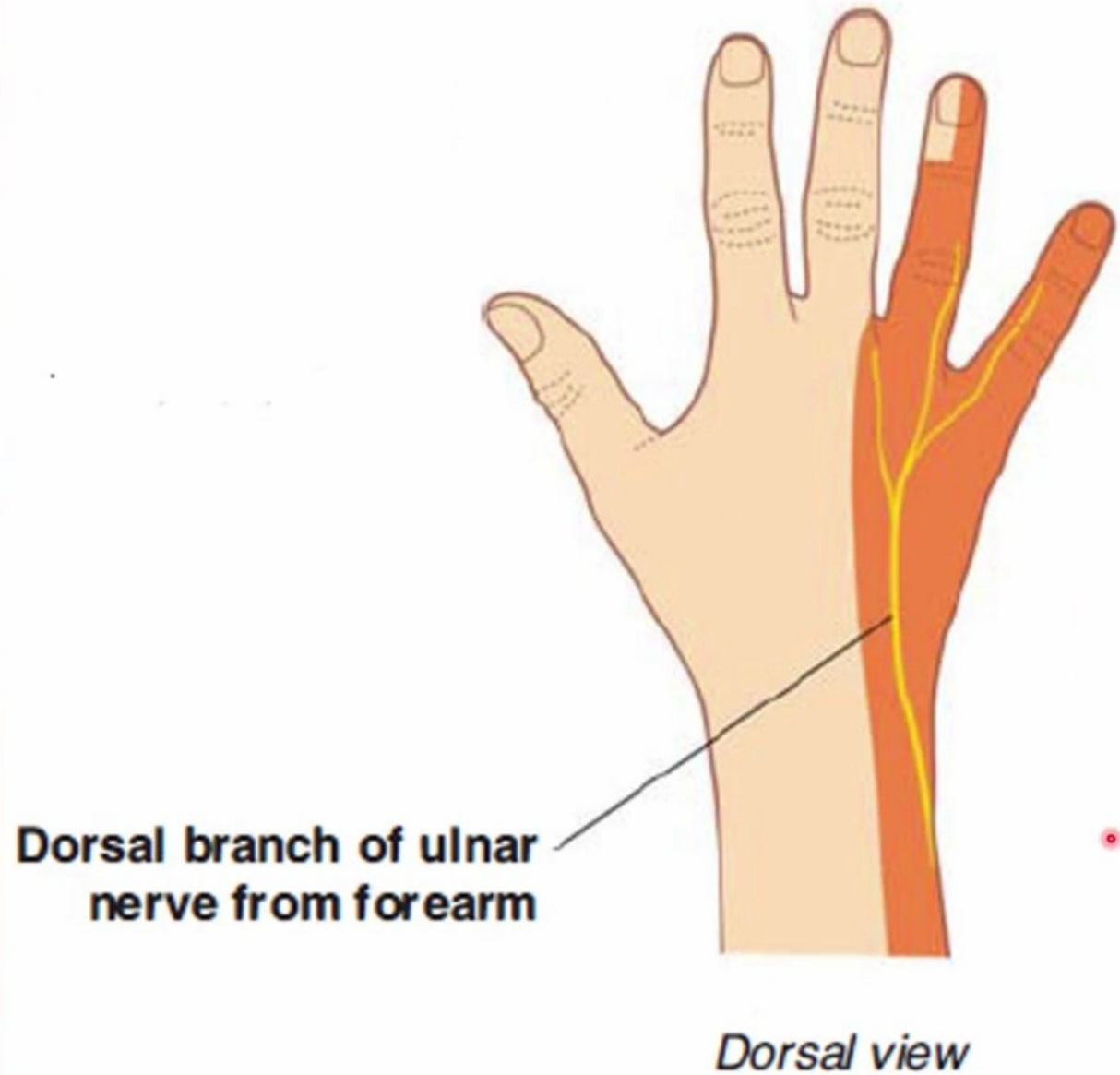
2) Deep terminal branch

- The deep branch passes deeply into **the palm** between the **heads of origin of flexor** and **abductor digiti minimi** and through the origin of **opponens digiti minimi**.
- Passing down to the **interossei**, it grooves the distal border of the **hook of the hamate** and **arches deeply in the palm** within the concavity of the **deep palmar arch**.
- It gives motor branches to:
 - The **three hypothenar muscles**.
 - The **two lumbricals** on the **ulnar side**.
 - **all the interossei**.
 - The **both heads of adductor pollicis**.



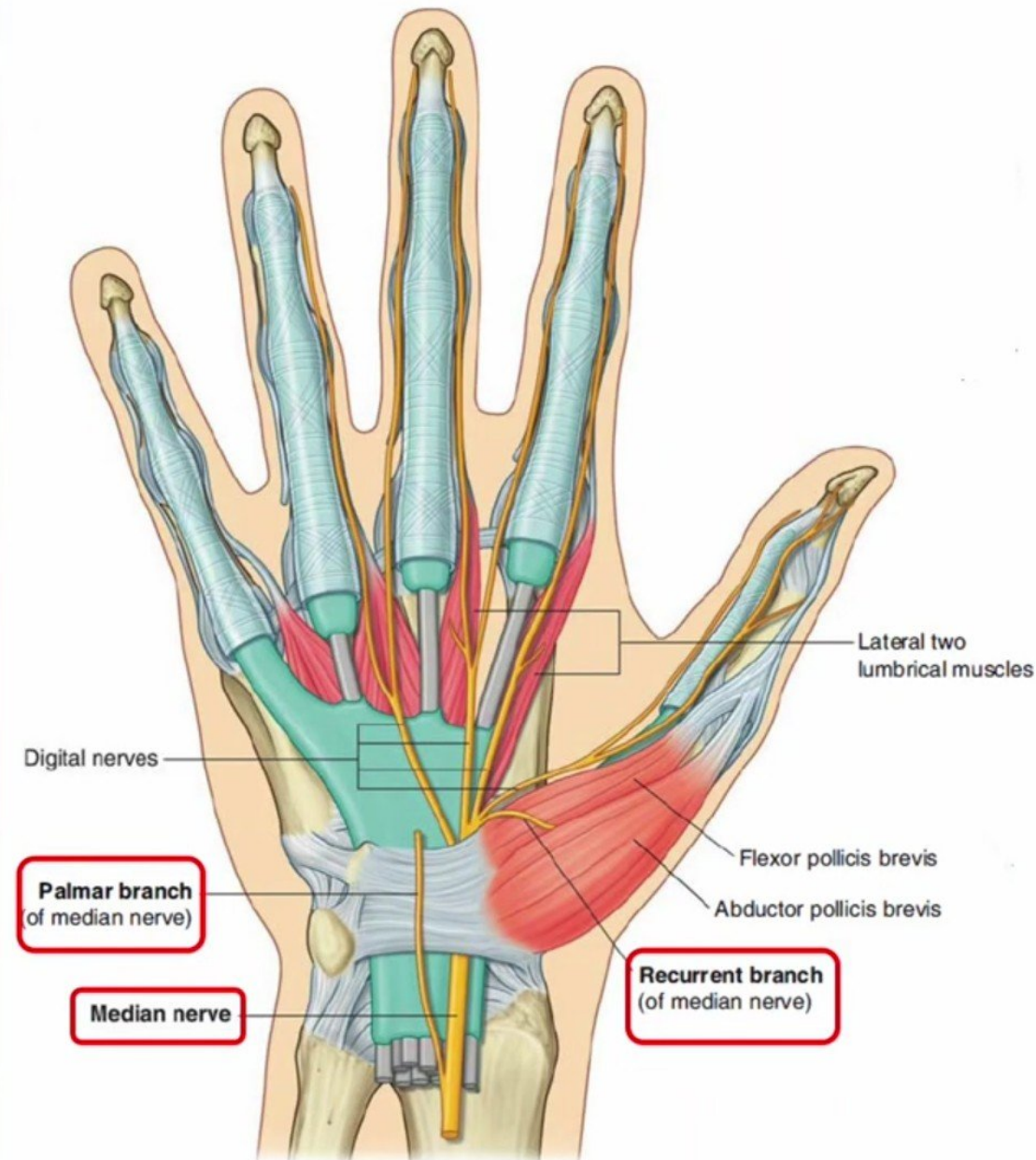
Dorsal branch of the ulnar nerve

- The **dorsal posterior cutaneous branch** is a **large branch** that arises in the **distal third of the forearm**.
- It passes medially between the tendon of the **flexor carpi ulnaris** and the **ulna** and
- It is distributed on the **posterior (dorsum) surface of the hand and fingers**.



The Median nerve of the hand

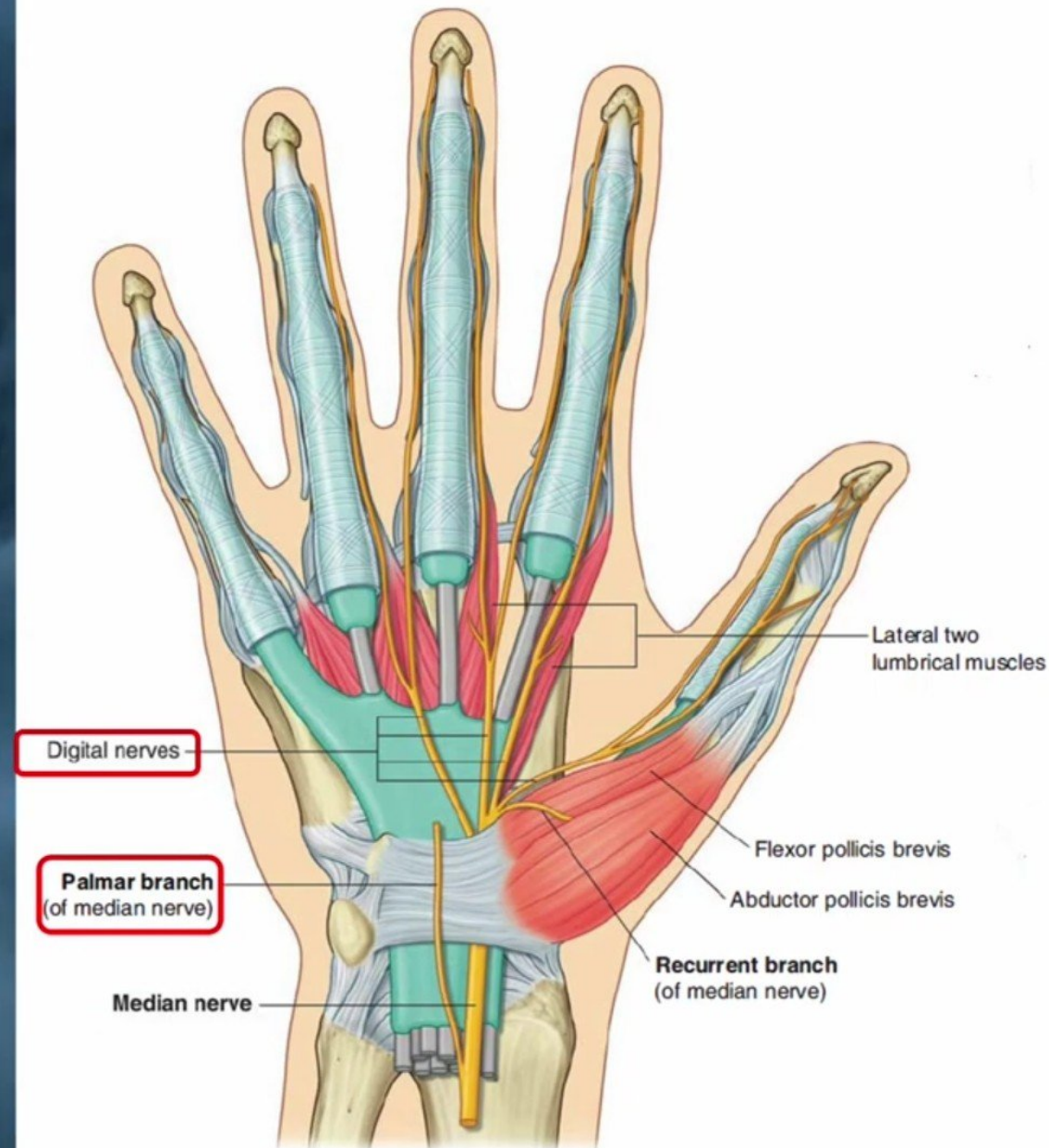
- It innervates skin by **sensory nerve** on the **thumb**, **index** and **middle** finger and lateral side of the ring finger ($3\frac{1}{2}$).
- It also innervates the thenar muscles. (opposition of the thumb to other digits).
- It enters the palm by passing behind the **flexor retinaculum** and through the carpal tunnel.
- It immediately divides into **lateral** and **medial branches**.
- **The muscular branch:**
 - It takes a recurrent course around the lower border of the flexor retinaculum.
 - It lies about **one fingerbreadth** distal to the tubercle of the scaphoid.
 - it supplies the muscles of the **thenar eminence** (the abductor pollicis brevis, the flexor pollicis brevis, and the opponens pollicis) and the 1st lumbrical muscle.



The Median nerve of the hand

The cutaneous (palmar digital) branches:

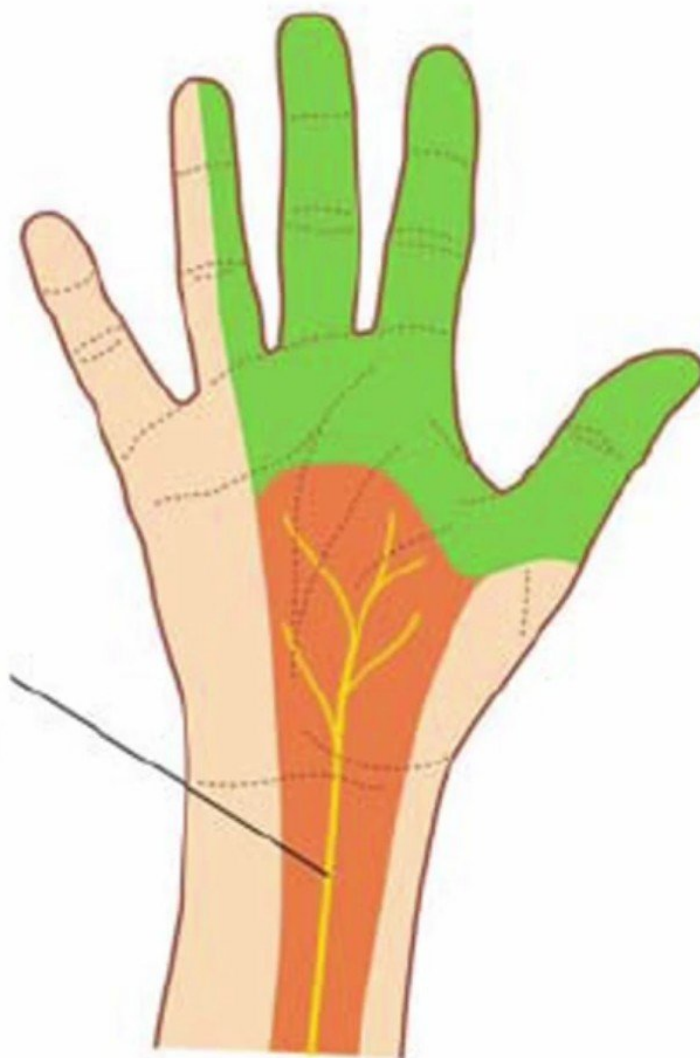
- **It supply:**
 - The palmar aspect of the lateral three and a half fingers.
 - The distal half of the dorsal aspect of each finger.
- One of these branches also supplies lumbrical muscles.
- **Note:**
 - The **palmar cutaneous branch** of the **median nerve** given off **in the front of the forearm crosses anterior to the flexor retinaculum** and supplies the skin over the lateral part of the palm.





Dorsal view

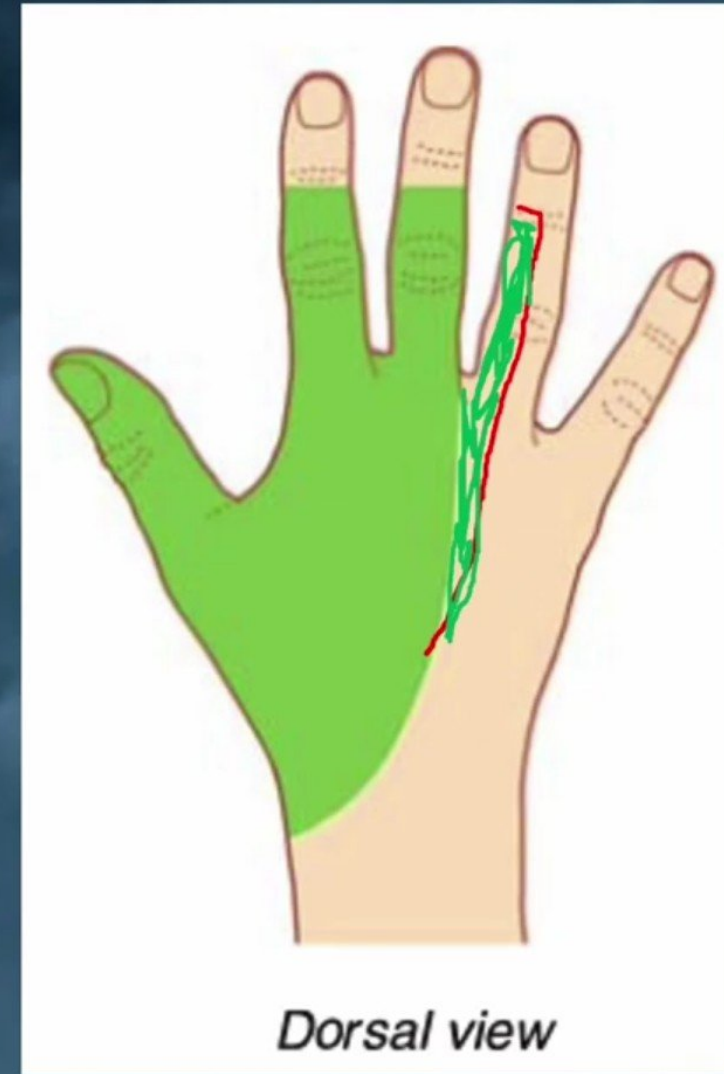
Palmar branch of median
nerve from forearm



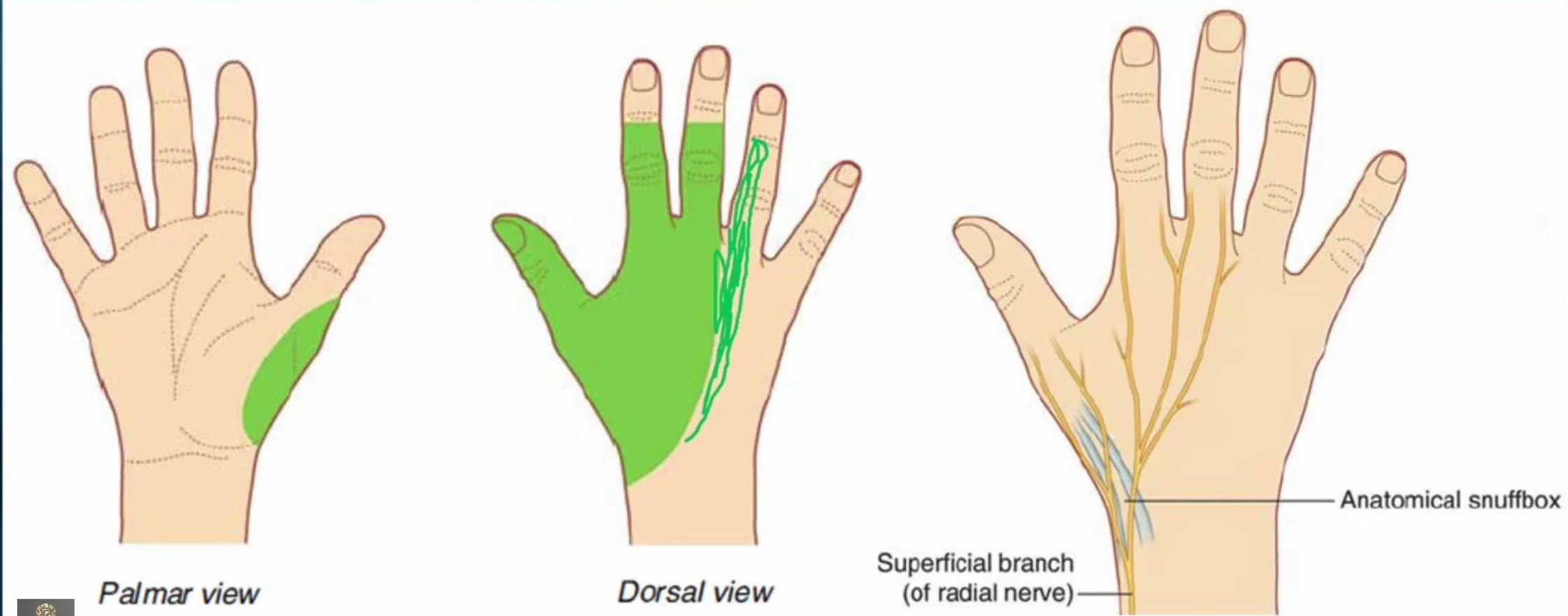
Palmar view

The Superficial Branch of the Radial Nerve of the hand

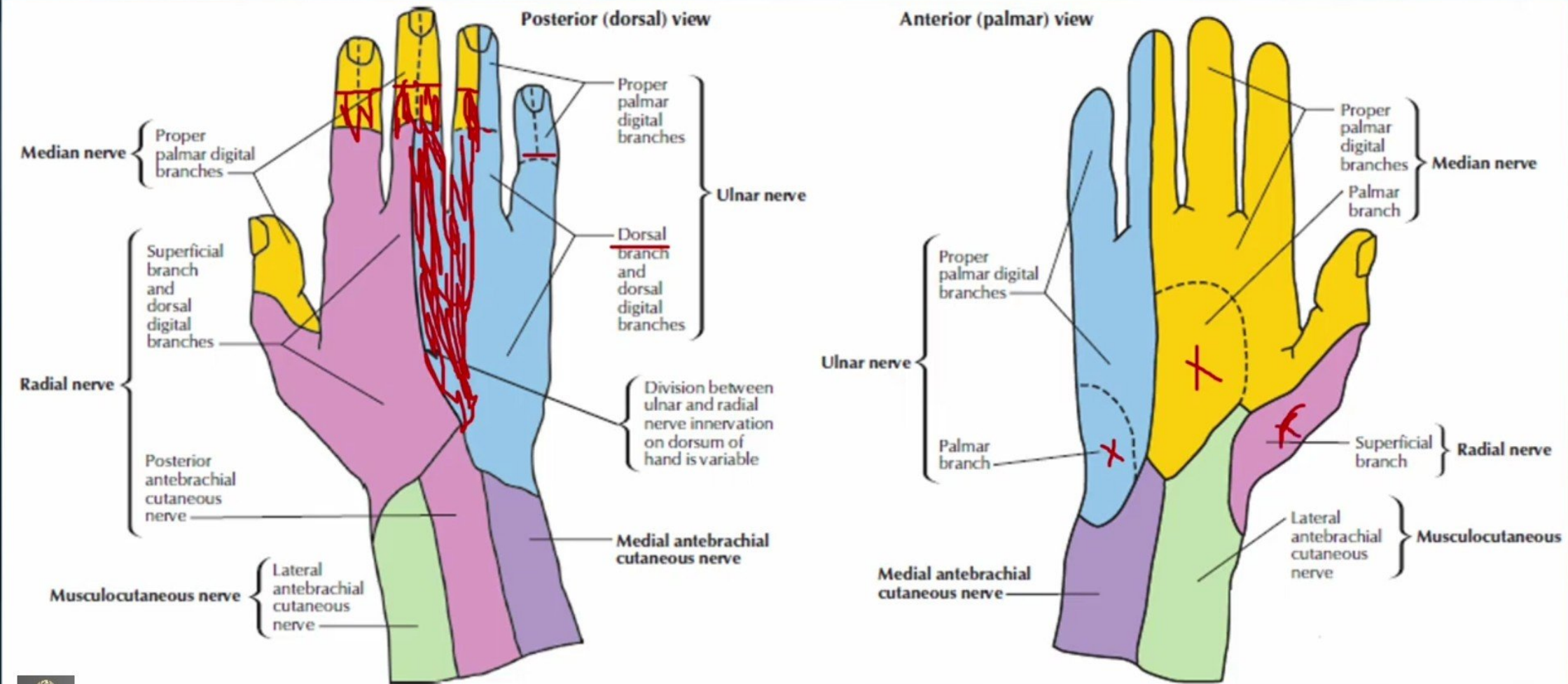
- It is **the direct continuation** of the nerve after its main stem has given off its deep branch **in front of the lateral epicondyle** of the humerus.
- It runs down under **cover of the brachioradialis muscle** on the **lateral side of the radial artery**.
- In the distal part of the forearm, it leaves the artery and passes backward under the tendon of the brachioradialis.
- It reaches the **posterior surface of the wrist**, where **it divides into** terminal branches that supply the skin on the **lateral two thirds of the posterior surface of the hand** and the posterior surface over the **proximal phalanges** of the **lateral three and a half fingers**.
- The area of skin supplied by the nerve on the dorsum of the hand is variable.



The radial innervation of the hand

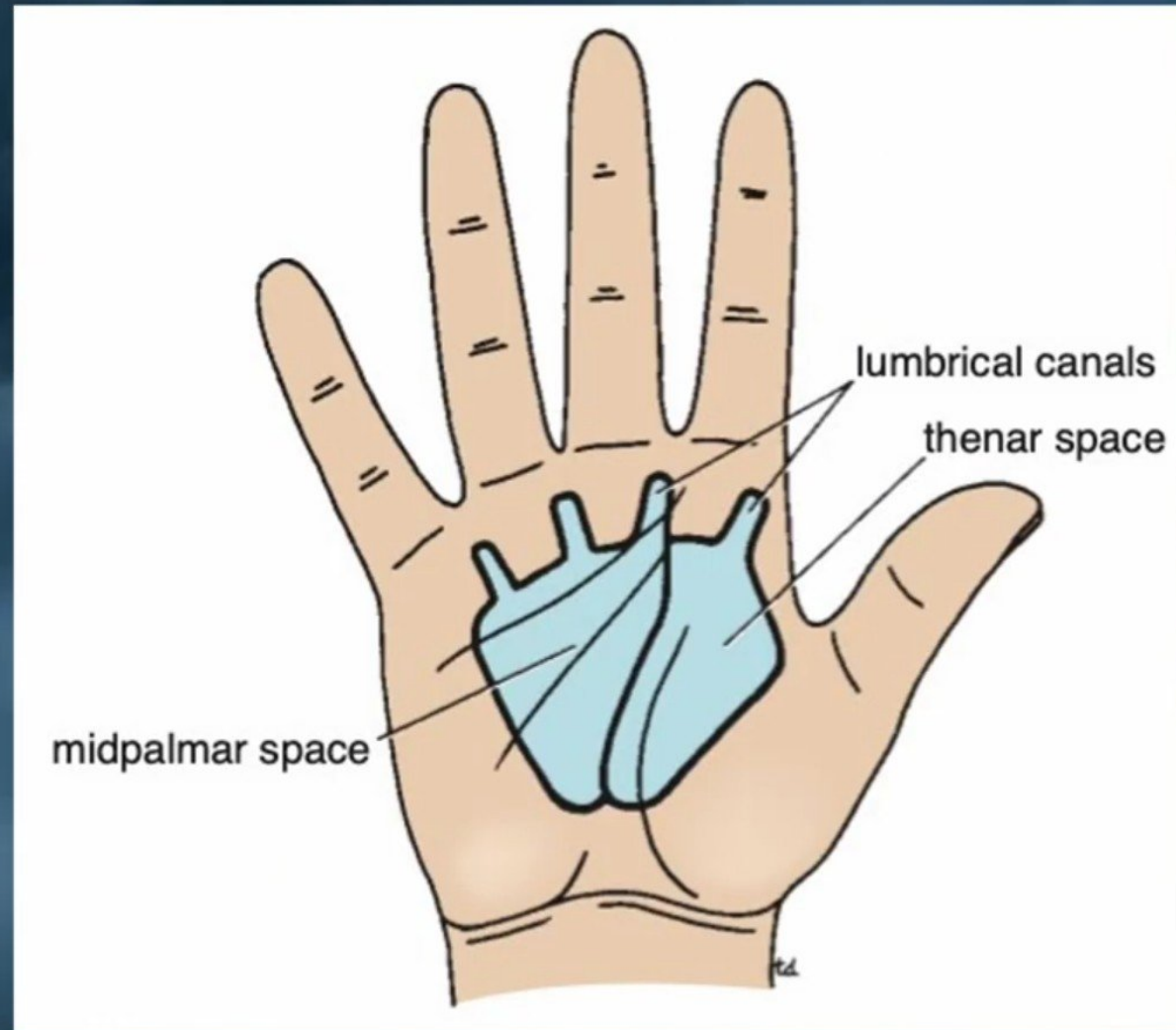


Summary of hand innervation



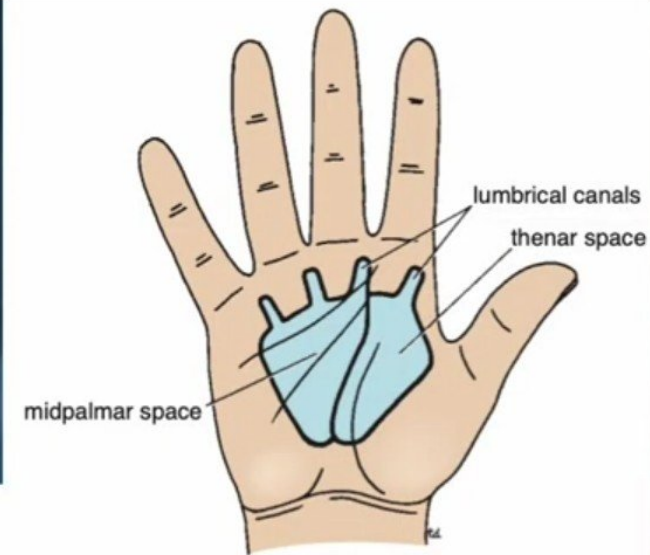
Fascial Palmer Spaces

- Normally, the fascial spaces of the palm are **potential spaces** filled with **loose connective tissue**.
- Their boundaries are important clinically because they may limit the spread of infection in the palm.

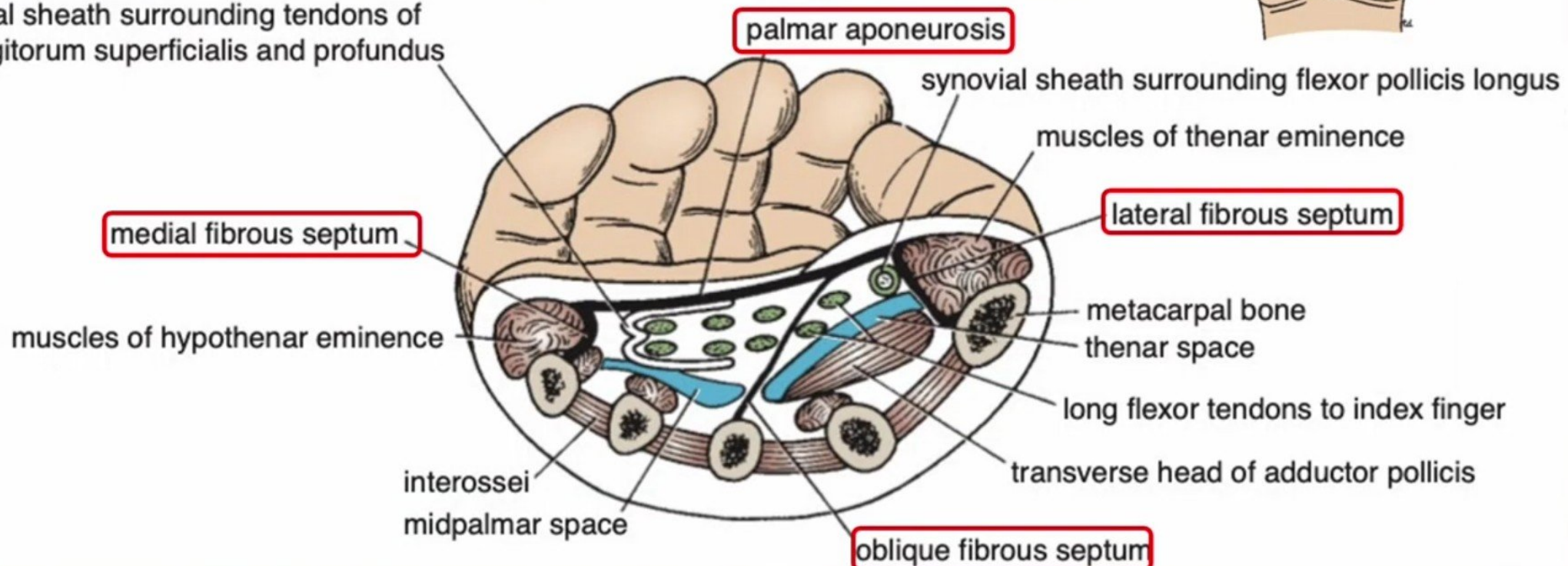


Fascial Palmer Spaces

- Palmar aponeurosis.
- Medial fibrous septum.
- Lateral fibrous septums
- Oblique fibrous septum.

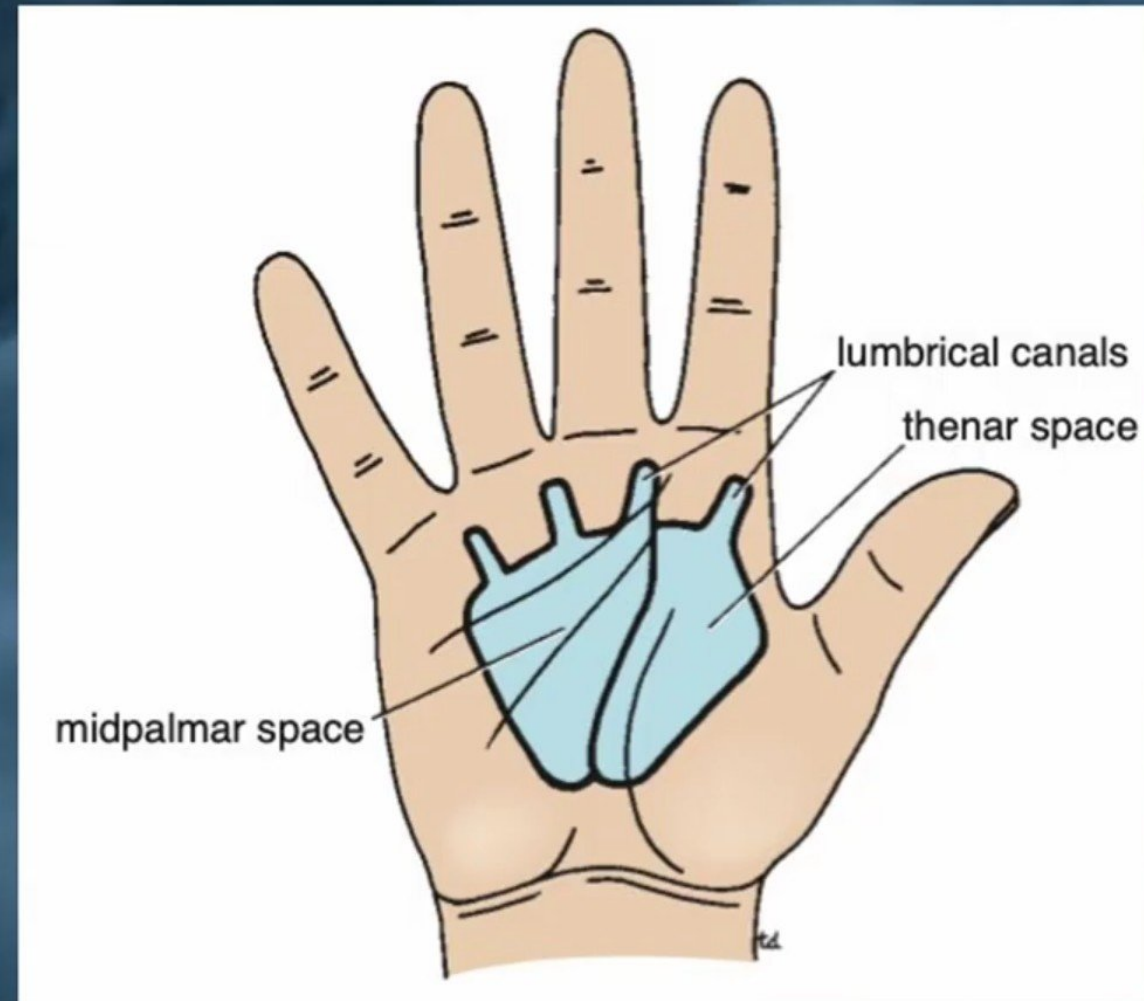


synovial sheath surrounding tendons of flexor digitorum superficialis and profundus



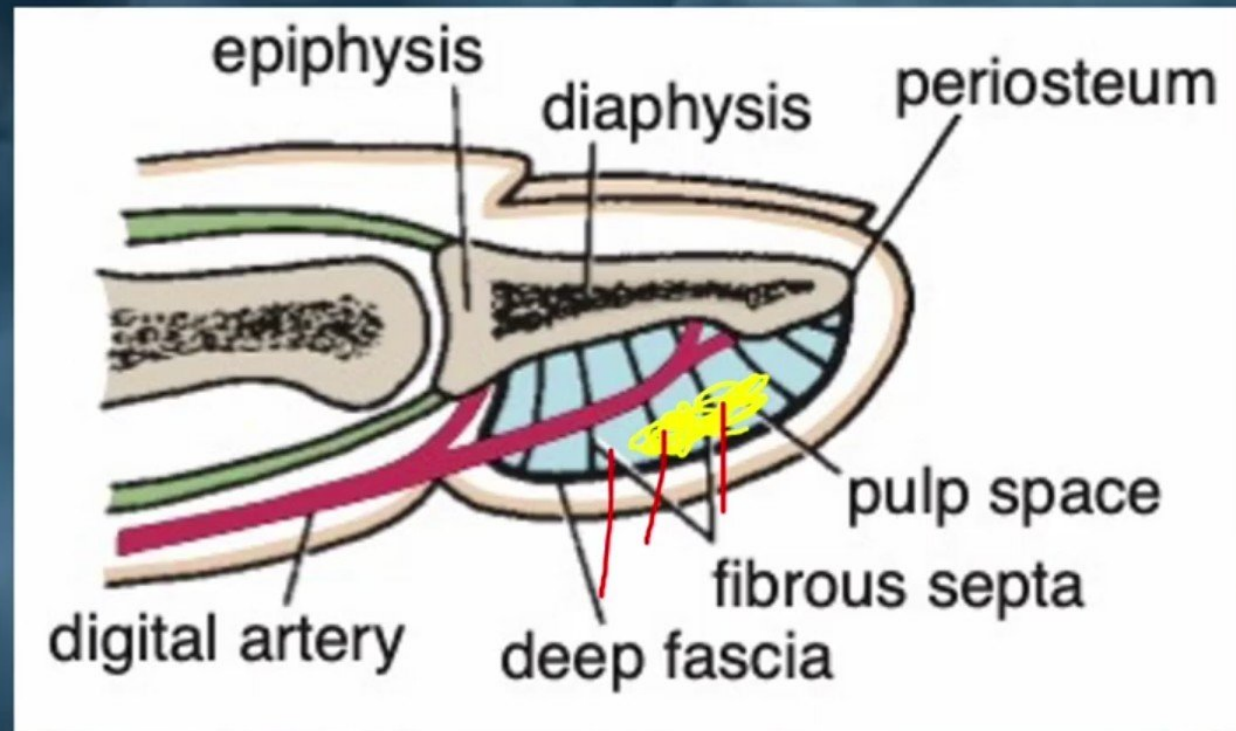
Fascial Palmer Spaces

- **The thenar space** contains:
 - ✓ **The first lumbrical muscle** (and lies posterior to the long flexor tendons to the index finger and in front of the adductor pollicis muscle).
- **The midpalmar space** contains:
 - ✓ **The 2nd, 3rd, and 4th lumbrical muscles** (and lies posterior to the long flexor tendons to the middle, ring, and little fingers. It lies in front of the interossei and the third, fourth, and fifth metacarpal bones.)
- **The lumbrical canal** is a **potential space** surrounding the tendon of each lumbrical muscle and is normally filled with connective tissue. Proximally, it is continuous with one of the palmar spaces.

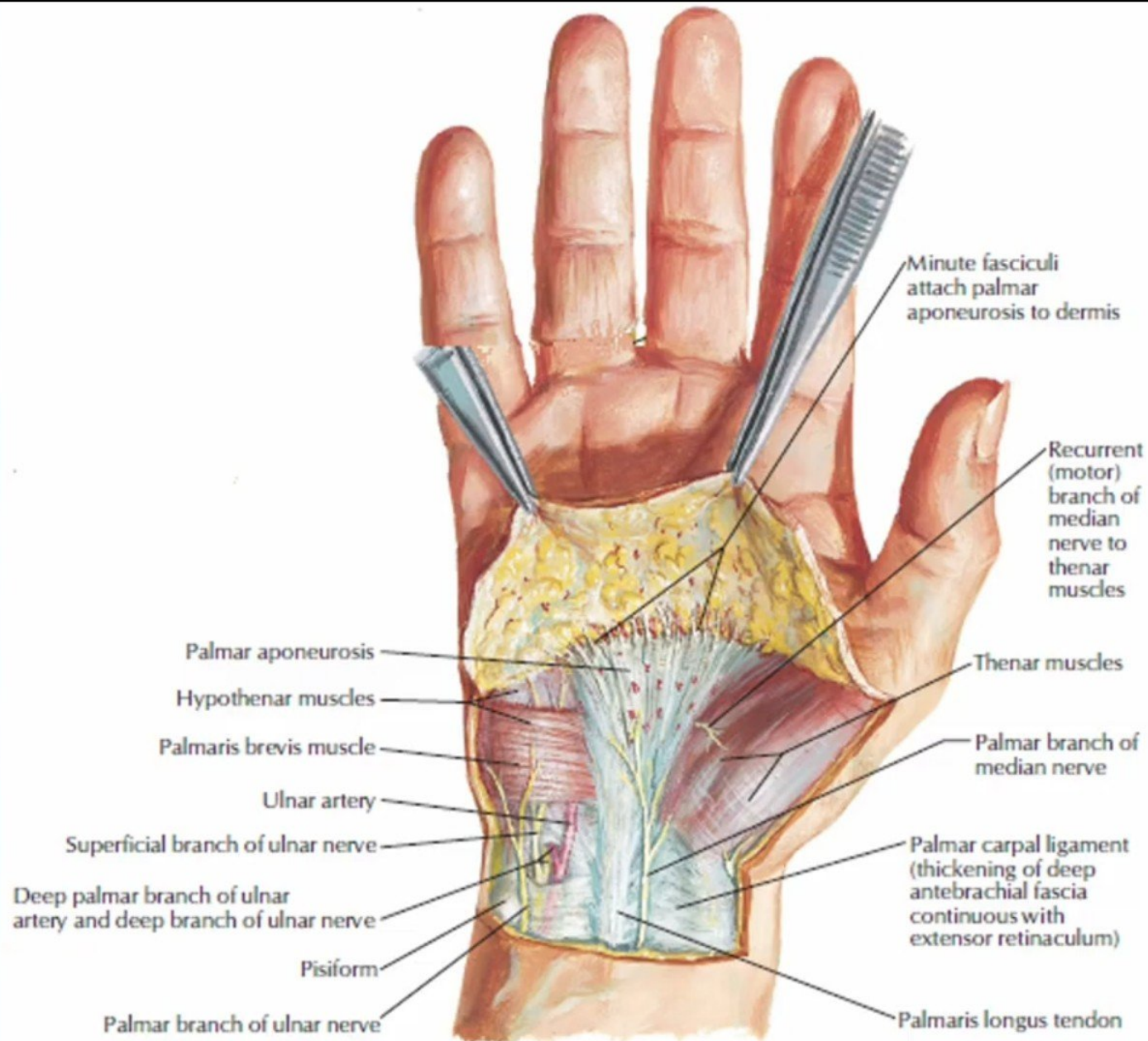


Pulp Space of the Fingers

- The **deep fascia of the pulp** of each finger fuses with **the periosteum of the terminal phalanx** just distal to the insertion of the long flexor tendons and **closes off a fascial compartment** known as the pulp space.
- Each pulp space is subdivided by the **presence of numerous septa**, which pass from the **deep fascia** to the periosteum.
- The **epiphysis** of the distal phalanx receives its **blood supply proximal** to the **pulp space**.

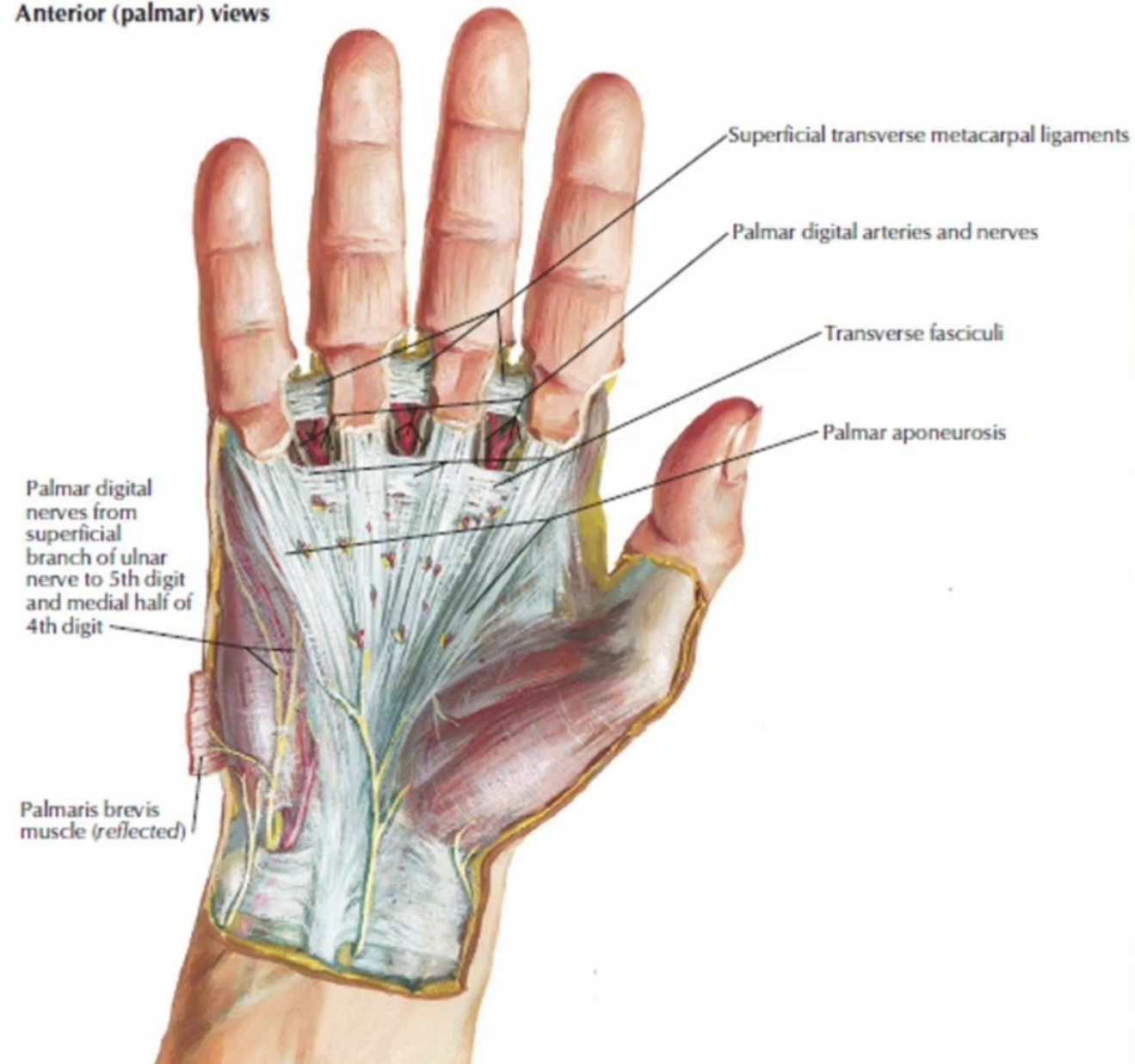


Hand Anatomical Dissection

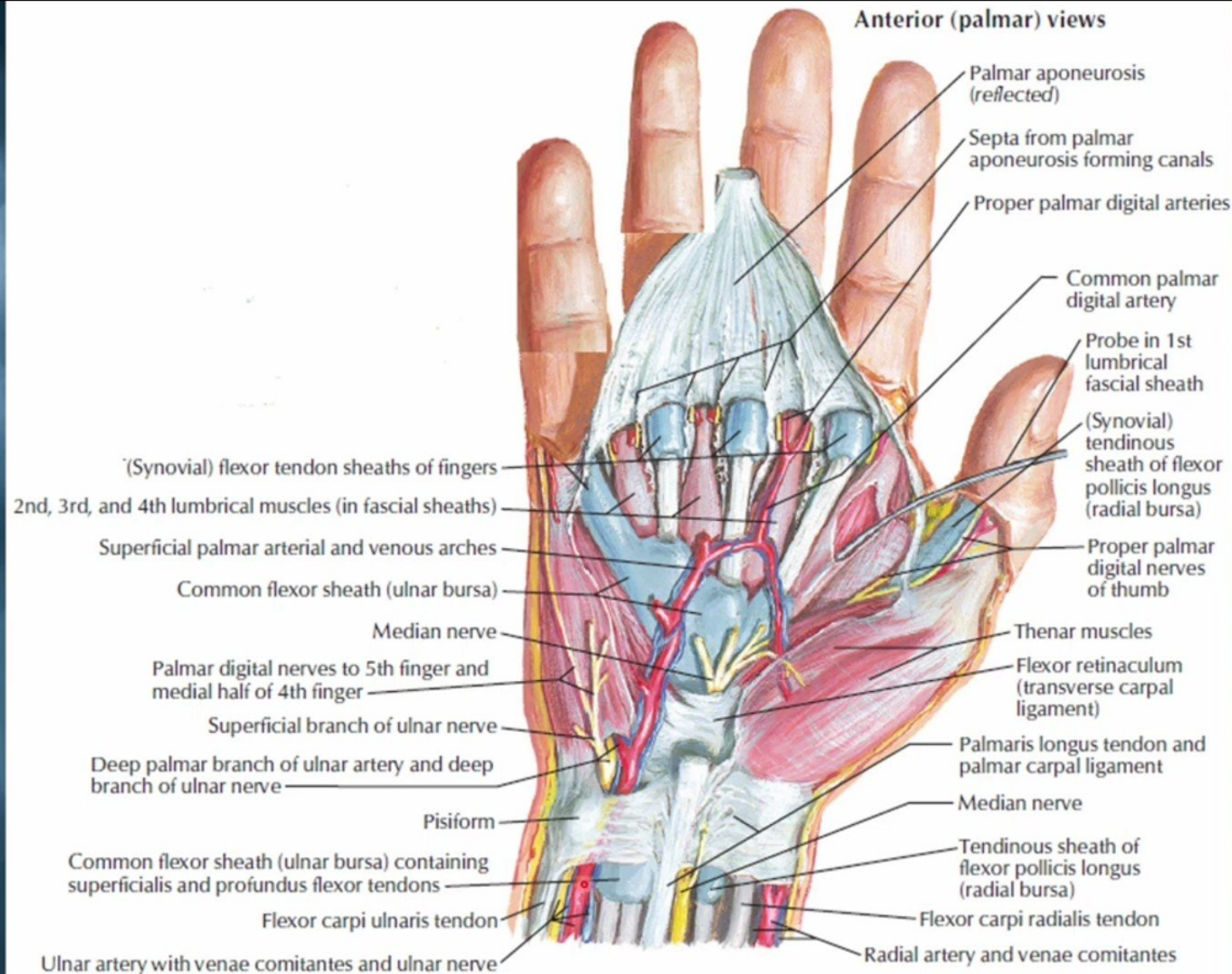


Hand Anatomical Dissection

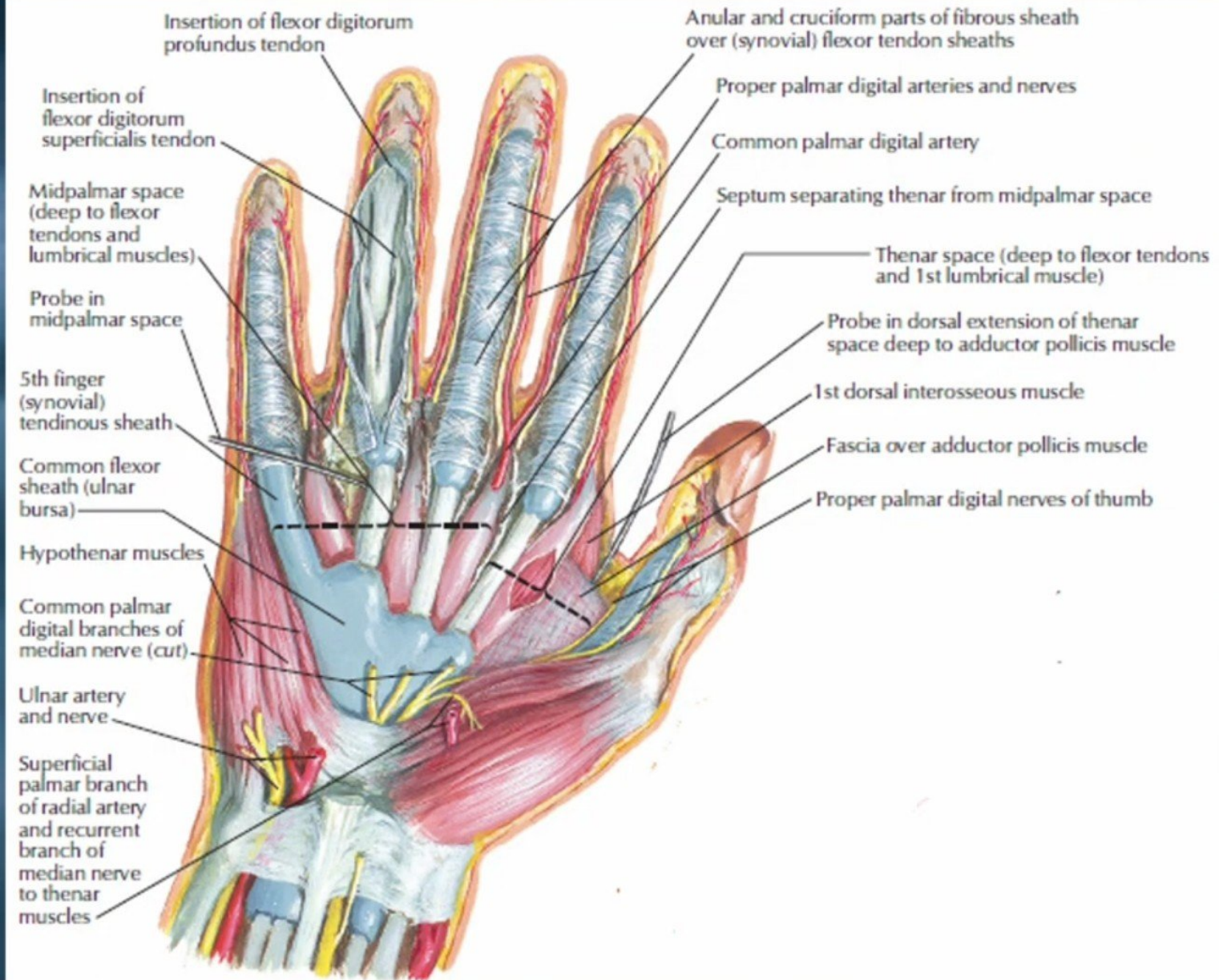
Anterior (palmar) views



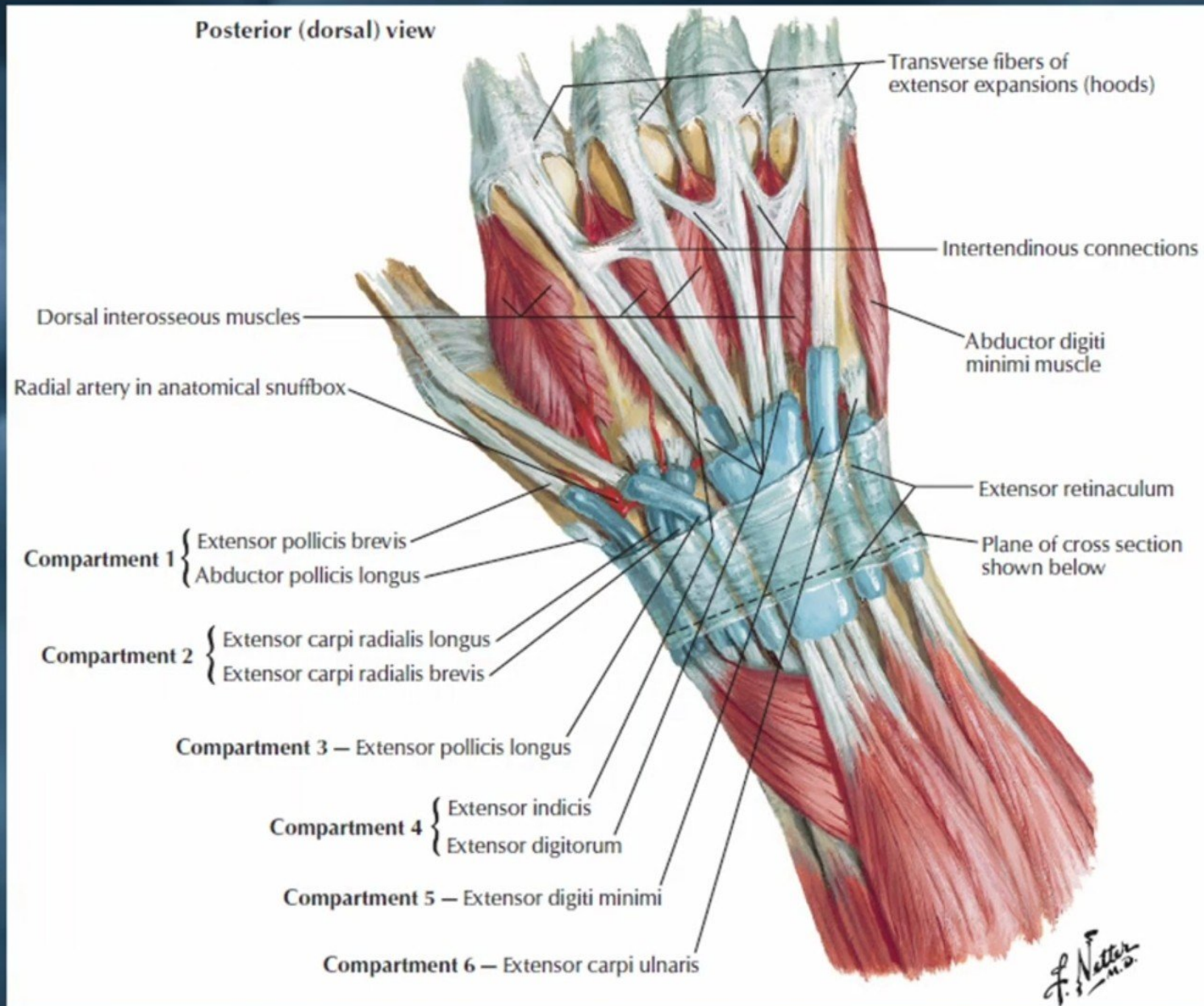
Hand Anatomical Dissection



Hand Anatomical Dissection



Hand Anatomical Dissection



Thanks